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NUMERICAL METHODS FOR POWER-LAW DIFFUSION PROBLEMS*

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Abstract. In this paper, we consider numerical methods for nonlinear diffusion problems where the diffusion term follows a power law, e.g., p-Laplace-type problems. In the first part, we present continuous higher order finite element discretizations for the model problem and we derive error estimates. In the second part, we discuss Newton iterative methods based on residual-based line-search and error-oriented globalization, which are employed for the numerical solution of the produced nonlinear algebraic system. Third, we formulate the original problem as a saddle point problem in the frame of augmented Lagrangian techniques and present two iterative methods for its solution. We conduct a systematic investigation of all solution algorithms. These algorithms are compared with respect to computational cost and their efficiency. Numerical results demonstrating the theoretical error estimates are also presented in five examples.

Key words. power-law diffusion problems, p-Laplace-type problems, high order finite element discretizations, Newton iterative methods, augmented Lagrangian techniques

AMS subject classifications. 65N12, 65N15, 65L20, 65L60, 49M15

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1. Introduction.

Second order elliptic problems with p-type gradient nonlinearities in the diffusion coefficient, that is, $(\varepsilon^2+|\nabla u|^2)^{\frac{p-2}{2}}$ with $\varepsilon \geq 0$ and $p \in (1,\infty)$, are of special interest and are relevant in many practical applications, e.g., in aerodynamics, porous media, non-Newtonian flows, plasticity and glaciology, and fluid-structure interaction [1, 2, 3, 4]. Therefore, the numerical solution of these problems is of great importance. In the application of numerical methods for solving these type of problems, we meet several difficulties associated mainly with the presence of the gradient in the p-nonlinear structure of the diffusion term.

The most classical p-type model problem that has been studied in the literature is the p-Laplace problem, i.e., $\varepsilon = 0$. The first analysis of finite element methods for p-Laplace was undertaken in [5] and in [6], (Chapter 5)], where (suboptimal) error estimates have been shown in the $W^{1,p}$-norm. These results were further improved in [7] and, recently, were used in [8, 9] for developing k and hk finite element methods, where h denotes the spatial discretization parameter and k the degree of the polynomial space.[1] Recently, optimal error bounds have been shown for p-Laplace and more general p-power-law elliptic problems by using quasi-norm interpolation estimates. The main idea in this approach is first to establish interpolation estimates in a quasi norm, which is predetermined by the p-nature of the problem, and then to derive error estimates by exploiting the relations between the quasi and Sobolev norms. Among others, we quote [10, 11, 12, 13]. In [14], interpolation operators in

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[1] In the literature, p usually denotes the degree of the polynomial space. Due to the appearance of the parameter p in the p-Laplace operator, we denote the polynomial degree by k.

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Orlicz–Sobolev spaces were studied and were utilized for approximating solutions of more general p-type problems, the so-called p-structure problems.

Over the last two decades, there has been an increasing interest in devising discontinuous Galerkin (DG) methods for the numerical solution of p-type problems; see [15, 16, 17, 18]. In all these DG methods the numerical fluxes were developed by following the p-nonlinear nature of the problem. Particularly, in [16], local DG methods were studied for p-structure problems. The quasi-norm interpolation estimates presented in [14], were applied in the frame of broken paces and optimal error estimates were shown for linear elements. The same local DG methods have been used later in [17] for solving more realistic p-type problems and in [19] for solving non-Newtonian flow problems.

We point out that the degenerate nature of the p-type problems makes the study of their regularity properties quite difficult. In general, we cannot guarantee high regularity for the solution u even with smooth problem data, [20] and, hence, the FE methods that have been applied for these problems usually consider first order polynomial spaces. In [21], global $W^{2,2}$ regularity properties were established under some assumptions on the right-hand side of the p-type model under consideration. These results were used in the discretization error analysis in [21, 22]. Furthermore, it has been shown in [22] that under some further assumptions on the problem data, we can obtain $W^{1+2/p,p}$ regularity for the solutions of the p-Laplacian, and this regularity is sufficient to ensure optimal error bounds.

However, recently there has been much work on applying high order tensor product spaces for solving elliptic problems, even in cases with reduced regularity, and many techniques have been developed, e.g., adaptive meshes, graded meshes, for recovering the optimal rates; see, for example, [8, 23, 24, 25, 26, 27]. In this paper, motivated by the results in [14], we discuss high order interpolation error estimates in the same quasi norms for solving p-type diffusion problems, with $p \in (1, \infty)$ and $0 < \varepsilon \leq 1$.[2]

In particular, assuming appropriate regularity properties for u (see Assumptions 2.1 and 2.6), we combine the structure (monotonicity) conditions presented in [14], with known high order interpolation estimates in Sobolev spaces, [28]. This helps us to derive quasi-norm interpolation error estimates using high order spaces. Having these estimates, we proceed and show optimal error bounds for sufficiently smooth solutions. We mention that in [14] quasi-norm error estimates have been shown for linear spaces but using a different methodology.

After the finite element discretization of the p-type diffusion problem we must solve a nonlinear algebraic system. In the second part of the paper, we present two Newton methods for solving the nonlinear algebraic system, (i) a classical residual-based procedure and (ii) an error-oriented approach [29]. We also mention that in [30], inexact Newton methods with adaptive stopping criteria for the nonlinear and the linear solver have been presented. Therein, the stopping criteria is based on a posteriori error estimates, which split the global error into three parts, the discretization, the linearization, and the algebraic.

Initially inspired by the results in [31], we show bounds for the condition number of the iterative Newton Jacobian matrix. Then we present two methods. The first

[2]We emphasize that in the numerical examples, we always use $\varepsilon^2 > 0$, which helps on relaxing the singular behavior of the problem at points where $|\nabla u| = 0$. However, we perform numerical tests setting $\varepsilon^2 = 10^{-8}$, which can give us an idea about the behavior of the algorithm for the classical p-Laplace problem.

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method is simple to be materialized and uses a classical criterion in order to check at each step whether the residual has been decreased. The second method is based on a natural monotonicity criterion and rather than observing the residuals, two update steps are compared. This second method is more expensive since an additional simplified problem needs to be solved. Later, in the numerical examples, we investigate and compare the performance and the corresponding convergence rates of the two Newton methods with respect to the variation of the problem parameters ε and p. The comparisons are made in a wide range of problems with high and low regularity solutions.

Last, we treat the original problem by developing techniques which provide a separation of the gradient of the solution from the nonlinearity. We achieve this by means of augmented Lagrangian (ALG) techniques. The original problem is reformulated as a saddle point problem where the produced variational problem is discretized by finite elements. We propose two iterative methods for solving the resulting nonlinear algebraic system. The first iterative method is the classical ALG1 iterative method, which is usually used in the literature; see a detailed analysis in [32]. It can be interpreted as a variant of the Uzawa algorithm, where the Lagrange multiplier is separately updated after a sequence of nested iteration procedure and is denoted by sALG1. The second proposed iterative method can be characterized as a monolithic approach, mALG1, where all the unknown variables (state variable and Lagrange multiplier) are simultaneously computed in one step. All proposed methods are compared with respect to computational cost and to the convergence rates in several examples. We point out that both ALG1 methods yield the same numerical solutions (in terms of accuracy) as the Newton methods. The mALG1 method is of optimal complexity and gives rise to a more easy handling of the p-nonlinear nature of the model, but is not as fast as the Newton methods. To the best of our knowledge, a comparison of these iterative techniques including also higher order finite elements discretizations has not yet been presented in the existing literature.

The outline of the paper is as follows. In section 2, the model problem is introduced. Furthermore, the finite element discretization analysis of this problem is presented. In section 3, residual-based and error-oriented Newton methods are formulated. Also, in the same section the solution of the problem under the ALG methodology is presented and the two ALG1 iterative methods are described in detail. Finally, in section 4 systematic comparisons of all solution algorithms in terms of convergence rates, Newton iterations, and CPU times are measured. We also present numerical results to confirm the theoretical error bounds. The paper closes with the conclusions in section 5.

2. The model problem. Let Ω be a bounded polygonal domain in $\mathbb{R}^d$, with $d = 2$ and $\Gamma_D = \partial\Omega$. We consider the following scalar p-type problem

$$-\mathrm{div}\mathbf{A}(\nabla u) = f \quad in\ \Omega, \qquad u = u_D \quad on\ \Gamma_D, \tag{2.1}$$

where $f : \Omega \to \mathbb{R}$ and $u_D : \Gamma_D \to \mathbb{R}$ are given smooth functions. The operator $\mathbf{A}(\nabla u) : \mathbb{R}^2 \to \mathbb{R}^2$ has the following p-power-law form

$$\mathbf{A}(\nabla u) = (\varepsilon^2 + |\nabla u|^2)^{\frac{p-2}{2}} \nabla u, \tag{2.2}$$

where $p \in (1, \infty)$ and $\varepsilon > 0$ are model parameters and $|\cdot|^2 = (\cdot, \cdot)$. The function $a(\nabla u) = (\varepsilon^2 + |\nabla u|^2)^{\frac{p-2}{2}}$ is the diffusivity term of (2.1). As we can observe by (2.2), the nonlinear nature of the problem is due to the appearance of $|\nabla u|$ in the

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diffusivity function and this poses numerical challenges. We introduce the closely related function $\mathbf{F}:\mathbb{R}^2\to\mathbb{R}^2$ to the operator $\mathbf{A}$ by

$$\mathbf{F}(\mathbf{a}) = (\varepsilon^2+|\mathbf{a}|^2)^{\frac{p-2}{4}}\mathbf{a}. \tag{2.3}$$

2.1. Preliminaries and notation. Let $1\le p\le\infty$ be fixed and l be a nonnegative integer. As usual, $L^p(\Omega)$ denotes Lebesgue spaces for which $\int_\Omega |u(x)|^p\,dx<\infty$, endowed with the norm $\|u\|_{L^p(\Omega)}=\big(\int_\Omega |u(x)|^p\,dx\big)^{\frac{1}{p}}$, and $W^{l,p}(\Omega)$ is the Sobolev space, which consists of the functions $u:\Omega\to\mathbb{R}$ such that their weak derivatives $D^\alpha u$ with $|\alpha|\le l$ belong to $L^p(\Omega)$. If $u\in W^{l,p}(\Omega)$, then its norm is defined by

$$\|u\|_{W^{l,p}(\Omega)}=\left(\sum_{0\le|\alpha|\le l}\|D^\alpha\phi\|^p_{L^p(\Omega)}\right)^{\frac{1}{p}} \quad\text{and}\quad \|u\|_{W^{l,\infty}(\Omega)}=\max_{0\le|\alpha|\le l}\|D^\alpha u\|_{L^\infty(\Omega)}$$

for $1\le p<\infty$ and $p=\infty$, respectively. We refer the reader to [33] for more details about Sobolev spaces. Furthermore, we define the spaces

$$W^{l,p}_D:=\{u\in W^{l,p}(\Omega): u|_{\partial\Omega}=u_D\} \quad\text{and}\quad W^{l,p}_0:=\{u\in W^{l,p}(\Omega): u|_{\partial\Omega}=0\}. \tag{2.4}$$

In what follows, positive constants c and C appearing in the inequalities are generic constants which do not depend on the mesh size h. We indicate on what the constants may depend for a better understanding of the proofs. Frequently, we will write $a\sim b$ meaning that $ca\le b\le Ca$ with c and C independent of the mesh size.

2.2. The weak problem. The weak formulation for (2.1) reads as follows: Find $u\in W^{1,p}_D$ such that

$$\begin{aligned} &B(u,\phi)=l_f(\phi)\ \forall\phi\in W^{1,p}_0(\Omega),\\ &\text{where}\quad B(u,\phi)=\int_\Omega \mathbf{A}(\nabla u)\cdot\nabla\phi\,dx \quad\text{and } l_f(\phi)=\int_\Omega f\phi\,dx. \end{aligned} \tag{2.5}$$

Depending on the form of $\mathbf{A}$ and on the range of p, the well-posedness has been examined by means of monotone operators in several works; see e.g., [6, 7].

Problem (2.5) is equivalent to the minimization problem

$$\text{find}\ \ u\in W^{1,p}_D \text{ such that } J(u)\le J(\phi)\ \forall\phi\in W^{1,p}_D, \tag{2.6}$$

where $J:W^{1,p}_D\to\mathbb{R}$ is defined by

$$J(\phi)=\frac{1}{p}\int_\Omega(\varepsilon^2+|\nabla\phi|^2)^{\frac{p}{2}}\,dx-\int_\Omega f\phi\,dx. \tag{2.7}$$

Furthermore, one can show that the Gateaux derivative of B is given by

$$\begin{aligned} B'(u)(v,w)=&\int_\Omega(\varepsilon^2+|\nabla u|^2)^{\frac{p-2}{2}}\nabla v\cdot\nabla w\,dx\\ &+(p-2)\int_\Omega(\varepsilon^2+|\nabla u|^2)^{\frac{p-4}{2}}(\nabla u\cdot\nabla v)(\nabla u\cdot\nabla w)\,dx \quad\text{for}\quad u,v,w\in W^{1,p}_D. \end{aligned} \tag{2.8}$$

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ASSUMPTION 2.1. *Let $l \geq 2$ be an integer and let $p \in (1,\infty)$ and $d = 2$. We assume that the solution u of (2.5) belongs to $V := W_D^{l,p}(\Omega)$, where either $(l-1)p > d$ and $p > 1$ or $(l-1)p < d$ and $p > \frac{2d}{d+1}$. Further, we assume that $\nabla \mathbf{F}(\nabla u) \in L^2(\Omega)$.*

For simplicity, we assume the following for the boundary data.

ASSUMPTION 2.2. *Let $u_D \in \mathbb{P}^k(\Gamma_D)$, where $\mathbb{P}^k$ is the space of polynomial degree less than or equal to k. Further, we assume that $\left| \int_{\Gamma_D} u_D \, ds \right| = 0$.*

2.3. Known inequalities. The following inequalities are going to be used in several places in the text. Hölder's and Young's inequalities read, for any ϵ, $0 < \epsilon < \infty$, and $1 < p, q < \infty$ such that $\frac{1}{p} + \frac{1}{q} = 1$, for $u \in L^p(\Omega)$ and $v \in L^q(\Omega)$, there holds

$$\left| \int_\Omega uv \, dx \right| \leq \|u\|_{L^p(\Omega)} \|v\|_{L^q(\Omega)}, \qquad \left| \int_\Omega uv \, dx \right| \leq \frac{\epsilon}{p} \|u\|_{L^p(\Omega)}^p + \frac{\epsilon^{-\frac{q}{p}}}{q} \|v\|_{L^q(\Omega)}^q. \tag{2.9a}$$

Poincaré–Friedrichs inequality (see [28]) reads for any $u \in W^{1,2}(\Omega)$, it holds

$$\|u\|_{L^2(\Omega)} \leq C(\Omega, \partial\Omega) \left(\|\nabla u\|_{L^2(\Omega)} + \left| \int_{\partial\Omega} u \, ds \right| \right). \tag{2.9b}$$

Next, we introduce some functions which will be useful to the following. For $p > 1$ and $\varepsilon > 0$, we define the functions

$$\widehat{\varphi}(t) := (\varepsilon + t)^{p-2}, \tag{2.10}$$

$$\varphi(t) := \int_0^t (\varepsilon^2 + s^2)^{\frac{p-2}{2}} s \, ds, \qquad \text{and it follows that } \ \varphi'(t) = (\varepsilon^2 + t^2)^{\frac{p-2}{2}} t. \tag{2.11}$$

By an easy computation we can show that $\varphi'(t) \sim t\varphi''(t)$ and $\varphi''(t) \sim \widehat{\varphi}(t)$.

LEMMA 2.3. *Let $\mathbf{A}$ be given by (2.2) and let $\mathbf{F}$ be defined by (2.3). Then the relations*

$$\big(\mathbf{A}(\mathbf{P}) - \mathbf{A}(\mathbf{Q})\big) \cdot (\mathbf{P} - \mathbf{Q}) \sim |\mathbf{F}(\mathbf{P}) - \mathbf{F}(\mathbf{Q})|^2 \tag{2.12a}$$

$$\sim \widehat{\varphi}(|\mathbf{P}| + |\mathbf{Q}|)|\mathbf{P} - \mathbf{Q}|^2, \tag{2.12b}$$

$$\big|\mathbf{A}(\mathbf{P}) - \mathbf{A}(\mathbf{Q})\big| \sim \widehat{\varphi}(|\mathbf{P}| + |\mathbf{Q}|)|\mathbf{P} - \mathbf{Q}|, \tag{2.12c}$$

holds for all $\mathbf{P}, \mathbf{Q} \in \mathbb{R}^2$.

Proof. The proofs are given in [34]. ∎

2.4. Finite element notation. Let $T_h = \{E_i\}_{i=1}^{N_E}$ be a quasi-uniform subdivision of Ω into quadrilateral elements (without hanging nodes) with the following properties: (i) $\overline{\Omega} = \cup_{i=1}^{N_E} \overline{E_i}$, (ii) if E_i, $E_j \in T_h$ then their closures are either disjoint, or have common vertex, or have common edge. As usual, we set h_{E_i} to be the diameter of $E_i \in T_h$ and the mesh size h is considered to be the maximum diameter of $E_i \in T_h$, i.e., $h := \max_{E_i \in T_h} h_{E_i}$.

Let $E \in T_h$ and $k \in \mathbb{N}$, we denote $\mathbb{Q}_k(E)$ by the space of tensor product polynomials on E of degree less than or equal to k in each variable. On T_h, we define the approximation spaces $V_{D,h}^{(k)} \subset V$, $V_{0,h}^{(k)}$ and $V_h^{(k)}$ as

$$V_{D,h}^{(k)} := \{\phi_h \in C(\overline{\Omega})\phi_h|_E \in \mathbb{Q}_k(E) \ \forall E \in T_h, \text{ and } \phi_h|_{\partial\Omega} = u_D\}, \tag{2.13a}$$

$$V_{0,h}^{(k)} := \{\phi_h \in C(\overline{\Omega})\phi_h|_E \in \mathbb{Q}_k(E) \ \forall E \in T_h, \text{ and } \phi_h|_{\partial\Omega} = 0\}, \tag{2.13b}$$

$$V_h^{(k)} := \{\phi_h \in C(\overline{\Omega})\phi_h|_E \in \mathbb{Q}_k(E) \ \forall E \in T_h\}. \tag{2.13c}$$

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LEMMA 2.4. *Let $\mathcal{I}_h^k$ be the corresponding global interpolation operator and let $u \in W^{l,p}(\Omega)$ with $k+1 \geq l \geq 2$ and $p > 1$. Then, there exists a constant $C_{intp} > 0$ independent of h such that the interpolation estimate,*

$$|u - \mathcal{I}_h^k u|_{W^{s,p}(\Omega)} \leq C_{intp} h^{l-s} |u|_{W^{l,p}(\Omega)}, \tag{2.14}$$

holds for $0 \leq s \leq l$.

Proof. The proof can be found in [28]. ∎

Remark 2.5. We present the error analysis for the spaces in (2.13) defined on quadrilateral elements, because these spaces are available in the software package deal.II [35, 36], which we use for performing the numerical tests. The same analysis can be applied for other spaces defined on other mesh elements, e.g. triangular elements.

We proceed to the discretization analysis under the following assumption.

ASSUMPTION 2.6. *Let $1 < p < 2$, $q = \frac{p}{p-1}$, and let $\tau = \varepsilon + |\nabla u| + |\nabla \mathcal{I}_h^k u|$. We assume $\frac{1}{\tau} \in L^q(\Omega)$.*

Remark 2.7. Note that, if $\tau \geq 1$, i.e., $\varepsilon = 1$, we have $\|\frac{1}{\tau}\|_{L^q(\Omega)} \leq \|\tau\|_{L^q(\Omega)}$. Now, we can mention two cases. If $u \in W^{l,p}$ with $l = 2$ and $2 > p > \frac{2d}{d+1}$ (for the $d = 2$ case that we study here, this means $2 > p > \frac{4}{3}$), then using the Sobolev inequality $\|u\|_{L^q(\Omega)} \leq \|u\|_{W^{1,p}(\Omega)}$, it is easy to see that Assumption 2.6 is fulfilled. Furthermore, if $u \in W^{l,p}$ with $l > 2$ and $(l-1)p - d > 0$, then again using the Sobolev inequality $\|u\|_{L^q(\Omega)} \leq \|u\|_{W^{1,p}(\Omega)}$, we can see that Assumption 2.6 is fulfilled.

In the analysis below, we will use some special cases of the Sobolev embeddings theorem; see [33]. Let $\hat{E}$ be the reference element, for example, the unit square, and let $\hat{u} \in W^{1+m,p}(\hat{E})$ with the integer $m \geq 1$ and $2 > p > 1$ such that either $m\,p > d$ and $p > 1$ or $m\,p < d$ and $p > \frac{2d}{d+1}$ (i.e., for $d = 2$, we have $m\,p > 2$ and $p > 1$ or $m\,p < 2$ and $p > \frac{4}{3}$). Then for $q > p$, it holds that

$$\|\hat{u}\|_{W^{1,q}(\hat{E})} \leq C \|\hat{u}\|_{W^{1+m,p}(\hat{E})}, \tag{2.15}$$

where the constant C depends on d, p, q, m, and $\hat{E}$. By the Sobolev embedding (2.15) we can derive that

$$\|\hat{u}\|_{W^{1,q}(\hat{E})} \leq C \left(\|\hat{u}\|^p_{L^p(\hat{E})} + |\hat{u}|^p_{W^{1,p}(\hat{E})} + |\hat{u}|^p_{W^{2,p}(\hat{E})} \right)^{\frac{1}{p}}. \tag{2.16}$$

Let $E \in T_h$ and let T_E be the unique affine transformation that maps the reference element $\hat{E}$ to $E \in T_h$,

$$T_E : \hat{E} \to E \quad \text{with} \quad \mathrm{T}_E(\hat{x}) = B\hat{x} + b, \tag{2.17}$$

where $|det(B)| = |E|$. By applying a change of variables and using scaling arguments, we can show that

$$|\hat{u}|_{W^{j,p}(\hat{E})} = h^{j - \frac{d}{p}} |u|_{W^{j,p}(E)} \quad \text{for} \quad 0 \leq j \leq l. \tag{2.18}$$

Finally, using (2.18) in (2.16), a simple computation yields that

$$|u|^2_{W^{1,q}(E)} \leq C h^{2(\frac{d}{q} - \frac{d}{p} - 1)} \left(\|u\|^p_{L^p(E)} + h^p |u|^p_{W^{1,p}(E)} + h^{2p} |u|^p_{W^{2,p}(E)} \right)^{\frac{2}{p}}. \tag{2.19}$$

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COROLLARY 2.8. *Let $u \in W^{1,q}(\Omega) \cap W^{l,p}(\Omega)$ with $q > p$, $l \geq 2$, and let $\mathcal{I}_h^k$ be the corresponding interpolation. Then by* (2.14) *and* (2.19)*, it follows that*

$$|u - \mathcal{I}_h^k u|^2_{W^{1,q}(E)} \leq C h^{2(\frac{d}{q} - \frac{d}{p} - 1)}\, h^{2\,l} \|u\|^2_{W^{2,p}(E)}. \tag{2.20}$$

LEMMA 2.9. *Let $\mathbf{A}$ be given by* (2.2) *and let $\mathbf{F}$ be defined by* (2.3)*. For $\delta > 0$, there exists a constant $c(\delta)$ such that for $u, v, w \in V$ the following relation holds:*

$$\begin{aligned}\int_\Omega \big(\mathbf{A}(\nabla u) - \mathbf{A}(\nabla v)\big) \cdot (\nabla w - \nabla v)\, dx \leq \delta \int_\Omega |\mathbf{F}(\nabla u) - \mathbf{F}(\nabla v)|^2\, dx \\ + c(\delta) \int_\Omega |\mathbf{F}(\nabla w) - \mathbf{F}(\nabla v)|^2\, dx.\end{aligned} \tag{2.21}$$

Proof. The proof can be found in [14]. ∎

2.5. Finite element approximation and error bounds. In this paragraph, we present the finite element discretization of (2.5). We develop our analysis inspired by the results in [14]. The finite element approximation of (2.5) reads as follows: Find $u_h \in V_{D,h}^{(k)}$ such that, for all $\phi_h \in V_{0,h}^{(k)}$, it holds that

$$B(u_h, \phi_h) = l_f(\phi_h). \tag{2.22}$$

PROPOSITION 2.10.

(i) *Let $\{a_i\}_{i=1}^N$ be a sequence of nonnegative numbers. If $p < q < \infty$ then*

$$\left(\sum_{i=1}^N a_i^p\right)^{\frac{1}{p}} \leq N^{\frac{1}{p} - \frac{1}{q}} \left(\sum_{i=1}^N a_i^q\right)^{\frac{1}{q}}. \tag{2.23}$$

(ii) *Let $1 < p \leq q < \infty$ and let $E \in T_h$. If $u \in L^q(E)$ then there exists a C depending on the quasi-uniformity properties of T_h such that*

$$\|u\|_{L^p(E)} \leq C h^{d(\frac{1}{p} - \frac{1}{q})} \|u\|_{L^q(E)}. \tag{2.24}$$

(iii) *Let $\{a_i\}_{i=1}^N$ be a sequence of nonnegative numbers. If $1 < p < q < \infty$ then*

$$\left(\sum_{i=1}^N a_i^q\right)^{\frac{1}{q}} \leq \left(\sum_{i=1}^N a_i^p\right)^{\frac{1}{p}}. \tag{2.25}$$

Proof.

(i) Let $p^* = \frac{q}{p} > 1$ and $q^* = \frac{p^*}{p^* - 1} = \frac{q}{q-p}$ its conjugate exponent. Recalling (2.9a) for vectors, we have

$$\left(\sum_{i=1}^N a_i^p\, 1\right) \leq \left(\sum_{i=1}^N a_i^q\right)^{\frac{p}{q}} \left(\sum_{i=1}^N 1\right)^{\frac{q-p}{q}}. \tag{2.26}$$

Taking the $\frac{1}{p}$th root in (2.26), we obtain (2.23).

(ii) Inequality (2.24) can be shown in the same way as inequality (2.23).

(iii) We observe that the function $f(x) = (a_1^x + a_2^x + \cdots + a_N^x)^{\frac{1}{x}}$ is decreasing for $x \geq 1$. Then inequality (2.25) follows. ∎

Below we prove the basic interpolation estimates in the related quasi norms.

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THEOREM 2.11. *Let u satisfy Assumptions* 2.1 *and* 2.2 *and let $\mathcal{I}_h^k u$ be the corresponding interpolant as defined above. Then, there exists $c \geq 0$ depending on ε, $|\nabla u|$, $|\nabla \mathcal{I}_h^k u|$, the constant in* (2.14), *and the constant in* (2.19), *but independent of the grid size h, such that the following estimate holds true:*

$$\int_\Omega |\mathbf{F}(\nabla u) - \mathbf{F}(\nabla \mathcal{I}_h^k u)|^2 \, dx \leq c\, h^{2(l-1)} \|u\|^2_{W^{l,p}(\Omega)}. \tag{2.27}$$

Proof. For simplicity in the formulas below, we introduce the notation $\tau := \widehat{\varphi}(|\nabla u| + |\nabla \mathcal{I}_h^k u|)|^{\frac{1}{p-2}} = \varepsilon + |\nabla u| + |\nabla \mathcal{I}_h^k u|$. By the relations (2.12), we conclude that

$$\int_\Omega |\mathbf{F}(\nabla u) - \mathbf{F}(\nabla \mathcal{I}_h^k u)|^2 \, dx \leq \int_\Omega \widehat{\varphi}(|\nabla u| + |\nabla \mathcal{I}_h^k u|)\, |\nabla u - \nabla \mathcal{I}_h^k u)|^2 \, dx. \tag{2.28}$$

Let us first prove the required relation for $p > 2$. Applying (2.9a) in (2.28) and setting $q = \frac{p}{2}$, we obtain

$$\begin{aligned} &\int_\Omega |\mathbf{F}(\nabla u) - \mathbf{F}(\nabla \mathcal{I}_h^k u)|^2 \, dx \\ &\quad \leq c \bigg(\int_\Omega |\nabla u - \nabla \mathcal{I}_h^k u|^{\frac{2p}{2}} \, dx \bigg)^{\frac{2}{p}} \bigg(\int_\Omega (\widehat{\varphi}(|\nabla u| + |\nabla \mathcal{I}_h^k u|)^{\frac{p}{p-2}} \, dx \bigg)^{\frac{p-2}{p}} \\ &\quad \leq c |\nabla u - \nabla \mathcal{I}_h^k u|^2_{W^{1,p}(\Omega)} \|\tau\|^{p-2}_{L^p(\Omega)} \leq c h^{2(l-1)} \|\tau\|^{p-2}_{L^p(\Omega)}, \end{aligned} \tag{2.29}$$

where the interpolation estimate (2.14) has been used.

For the case $p < 2$ we proceed as follows. Let $E \in T_h(\Omega)$ and let $q^* = \frac{q}{2}$ and $p^* = \frac{p}{2-p}$. By relations (2.12), we can conclude that

$$\int_E |\mathbf{F}(\nabla u) - \mathbf{F}(\nabla \mathcal{I}_h^k u)|^2 \, dx \leq \int_E \widehat{\varphi}(|\nabla u| + |\nabla \mathcal{I}_h^k u|)\, |\nabla u - \nabla \mathcal{I}_h^k u)|^2 \, dx. \tag{2.30}$$

We apply (2.9a) in (2.30) and, consequently (2.19), and obtain

$$\begin{aligned} &\sum_E \int_E |\mathbf{F}(\nabla u) - \mathbf{F}(\nabla \mathcal{I}_h^k u)|^2 \, dx \\ &\quad \leq c \sum_E \bigg(\int_E |\nabla u - \nabla \mathcal{I}_h^k u|^{\frac{2q}{2}} \, dx \bigg)^{\frac{2}{q}} \bigg(\int_E \tau^{-p} \, dx \bigg)^{\frac{2-p}{p}} \\ &\quad \leq c \sum_E |u - \mathcal{I}_h^k u|^2_{W^{1,q}(E)} \bigg\| \frac{1}{\tau} \bigg\|^{2-p}_{L^p(E)} \\ &\quad \leq c \bigg(\sum_E |u - \mathcal{I}_h^k u|^q_{W^{1,q}(E)} \bigg)^{\frac{2}{q}} \bigg(\sum_E \bigg\| \frac{1}{\tau} \bigg\|^p_{L^p(E)} \bigg)^{\frac{2-p}{p}} \\ &\quad \leq c \bigg(\sum_E |u - \mathcal{I}_h^k u|^2_{W^{1,q}(E)} \bigg)^{\frac{2}{2}} \bigg(\sum_E \bigg\| \frac{1}{\tau} \bigg\|^p_{L^p(E)} \bigg)^{\frac{2}{p}} \bigg(\sum_E \bigg\| \frac{1}{\tau} \bigg\|^p_{L^p(E)} \bigg)^{-1} \end{aligned} \tag{2.31}$$

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$$\leq c\,h^{2(\frac{d}{q}-\frac{d}{p}-1)}\,h^{2\,l}\|u\|^2_{W^{2,p}(\Omega)}\,h^{2d(\frac{1}{p}-\frac{1}{q})}\left(\sum_E\left\|\frac{1}{\tau}\right\|^p_{L^q(E)}\right)^{\frac{2}{p}}\left(\left\|\frac{1}{\tau}\right\|^p_{L^p(\Omega)}\right)^{-1}$$

$$\leq ch^{2l-2}\|u\|^2_{W^{2,p}(\Omega)}\left(\sum_E\left\|\frac{1}{\tau}\right\|^p_{L^q(E)}\right)^{\frac{2}{p}}\left(\left\|\frac{1}{\tau}\right\|^p_{L^p(\Omega)}\right)^{-1},$$

where the interpolation estimate (2.14) and inequalities in Proposition 2.10 have been used. Finally, putting together the results (2.29) and (2.31), we have that

$$\int_\Omega |\mathbf{F}(\nabla u)-\mathbf{F}(\nabla\mathcal{I}_h^k u)|^2\,dx \leq c\,h^{2l-2}\|u\|^2_{W^{l,p}(\Omega)}. \tag{2.32}$$

This proves the theorem. ∎

Let us now use the previous results for showing an estimate for the approximation error $u_h - u$.

THEOREM 2.12. *Let $u \in V$ be the solution of* (2.5) *under Assumption* 2.1*, and let $u_h \in V^{(k)}_{D,h}$ be the solution of* (2.22)*. Then, there exists $C \geq 0$ depending on the constant in* (2.27) *but independent of the grid size h, such that*

$$\int_\Omega |\mathbf{F}(\nabla u)-\mathbf{F}(\nabla u_h)|^2\,dx \leq Ch^{2(l-1)}\|u\|^2_{W^{l,p}(\Omega)}. \tag{2.33}$$

Proof. Let $\phi_h \in V^{(k)}_{0,h}$ and let $\mathcal{I}_h^k u \in V^{(k)}_{D,h}$ be the interpolant of u. By forms (2.5) and (2.22), we can deduce that

$$\int_\Omega \big(\mathbf{A}(\nabla u)-\mathbf{A}(\nabla u_h)\big)\cdot\nabla\phi_h\,dx = 0, \tag{2.34}$$

where it follows that

$$\int_\Omega \big(\mathbf{A}(\nabla u_h)-\mathbf{A}(\nabla\mathcal{I}_h^k u)\big)\cdot\nabla\phi_h\,dx = \int_\Omega \big(\mathbf{A}(\nabla u)-\mathbf{A}(\nabla\mathcal{I}_h^k u)\big)\cdot\nabla\phi_h\,dx. \tag{2.35}$$

Now, choosing $\phi_h = u_h - \mathcal{I}_h^k u$ and $\delta > 0$ small enough, relations (2.12) and inequality (2.21) imply that

$$\int_\Omega |\mathbf{F}(\nabla u_h)-\mathbf{F}(\nabla\mathcal{I}_h^k u)|^2\,dx \leq c(\delta)\int_\Omega |\mathbf{F}(\nabla u)-\mathbf{F}(\nabla\mathcal{I}_h^k u)|^2\,dx, \tag{2.36}$$

where by the triangle inequality it follows that

$$\int_\Omega |\mathbf{F}(\nabla u_h)-\mathbf{F}(\nabla u)|^2\,dx \leq c(\delta)\int_\Omega \big|\mathbf{F}(\nabla u)-\mathbf{F}(\nabla\mathcal{I}_h^k u)|^2\,dx. \tag{2.37}$$

The desired estimate (2.33) follows immediately by applying the interpolation estimate (2.27) on the right-hand side of (2.37). ∎

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3. Nonlinear iterative processes. In this section, we design several iterative methods for solving problem (2.22). All proposed schemes are based on Newton-type approaches. We begin with an inspection of the behavior of the eigenvalues of the Jacobian matrix of $\mathbf{A}$ (see (2.2)), in order to get a preview of the behavior of the corresponding Newton iterative matrix around the critical points. Then, in section 3.3, two Newton-type iterative methods, e.g., residual-based descent and error-oriented techniques, are formulated. In section 3.4, ALG-type techniques are employed in order to reformulate problem (2.1) as a saddle point problem. We discretize this last problem using a classical finite element procedure and solve the resulting nonlinear system by applying two iterative procedures. Our final goal is to investigate the performance of all the aforementioned methods through the numerical examples.

3.1. Preliminary considerations. In view of the form of operator $\mathbf{A}$ in (2.2), we introduce the function $\mathbf{A} \in C^1(\mathbb{R}^2, \mathbb{R}^2)$ defined by

$$\mathbf{A}(\eta) = (\varepsilon^2 + \eta_1^2 + \eta_2^2)^{\frac{p-2}{2}}(\eta_1, \eta_2). \tag{3.1}$$

By an easy computation, we can show that

$$J_{\mathbf{A}} = \begin{bmatrix} (\eta_1^2+\eta_2^2+\varepsilon^2)^{(p-4)/2}((p-1)\eta_1^2+\eta_2^2+\varepsilon^2) & (p-2)\eta_1\eta_2(\eta_1^2+\eta_2^2+\varepsilon^2)^{(p-4)/2} \\ (p-2)\eta_1\eta_2(\eta_1^2+\eta_2^2+\varepsilon^2)^{(p-4)/2} & (\eta_1^2+\eta_2^2+\varepsilon^2)^{(p-4)/2}((p-1)\eta_2^2+\eta_1^2+\varepsilon^2)) \end{bmatrix} \tag{3.2}$$

with the eigenvalues

$$\lambda_1 = (\varepsilon^2 + \eta_1^2 + \eta_2^2)^{(p-2)/2}, \tag{3.3a}$$

$$\lambda_2 = (\varepsilon^2 + \eta_1^2 + \eta_2^2)^{(p-4)/2}\big((p-1)(\eta_1^2+\eta_2^2) + \varepsilon^2\big). \tag{3.3b}$$

The behavior of the eigenvalues of the Jacobian $J_{\mathbf{A}} := \frac{\partial \mathbf{A}(\eta)}{\partial \eta}$ with respect to the parameters ε and p, outlines the behavior of the eigenvalues of the corresponding Jacobian matrix, which appears in the Newton iterative process. Namely, for the cases where the eigenvalues of $J_{\mathbf{A}}$ are close to zero, we can expect the Jacobian matrix of the Newton method to be singular or ill-conditioned. Note that the eigenvalues λ_i, $i = 1, 2$, in (3.3) have a similar form to the function $\widehat{\varphi}$ defined in (2.10). For example, if we set $t = |\eta|$ then $\lambda_i(t) \sim \widehat{\varphi}(t)$, $i = 1, 2$.

Let $r = (\eta_1, \eta_2)$ be the radial function with $r^2 \leq 1$. In Figure 1, we plot λ_1 and λ_2 with respect to $r^2 = \eta_1^2 + \eta_2^2$ for several values of ε and p. We observe that for $p < 2$ and $\varepsilon < 1$ the values of both eigenvalues are getting very high as $r^2 \to 0$ and the values are reduced as $r^2 \to 1$; see Figures 1(a),(b),(d),(e). Conversely for the case where $\varepsilon = 1$ the eigenvalues are close to one; see Figures 1(c),(f). Concerning the graphs for $p > 2$, we can see that for $\varepsilon < 1$ the eigenvalues are close to zero as $r^2 \to 0$, and are increasing rapidly as $r^2 \to 1$; see Figures 1(g),(h). On the other hand, for $\varepsilon = 1$ the eigenvalues are greater than one for all r^2 values; see Figure 1(i).

Remark 3.1. For simplicity the analysis above has been presented in $\mathbb{R}^2$. It can be easily extended to $\mathbb{R}^{N_h}$, where N_h is the dimension of $V_{D,h}^{(k)}$.

3.2. The nonlinear algebraic system. Let N_h be the dimension of the space $V_{D,h}^{(k)}$ and $\{\phi_{h,i}\}_{i=1}^{N_h}$ are the basis functions of $V_{D,h}^{(k)}$. The solution $u_h \in V_{D,h}^{(k)}$ of problem (2.22) is expressed as $u_h = \sum_i^{N_h} U_i \phi_{h,i}(x)$, where the U_i are the degrees of freedom (DoFs). When this expression is substituted into (2.22), we obtain the following

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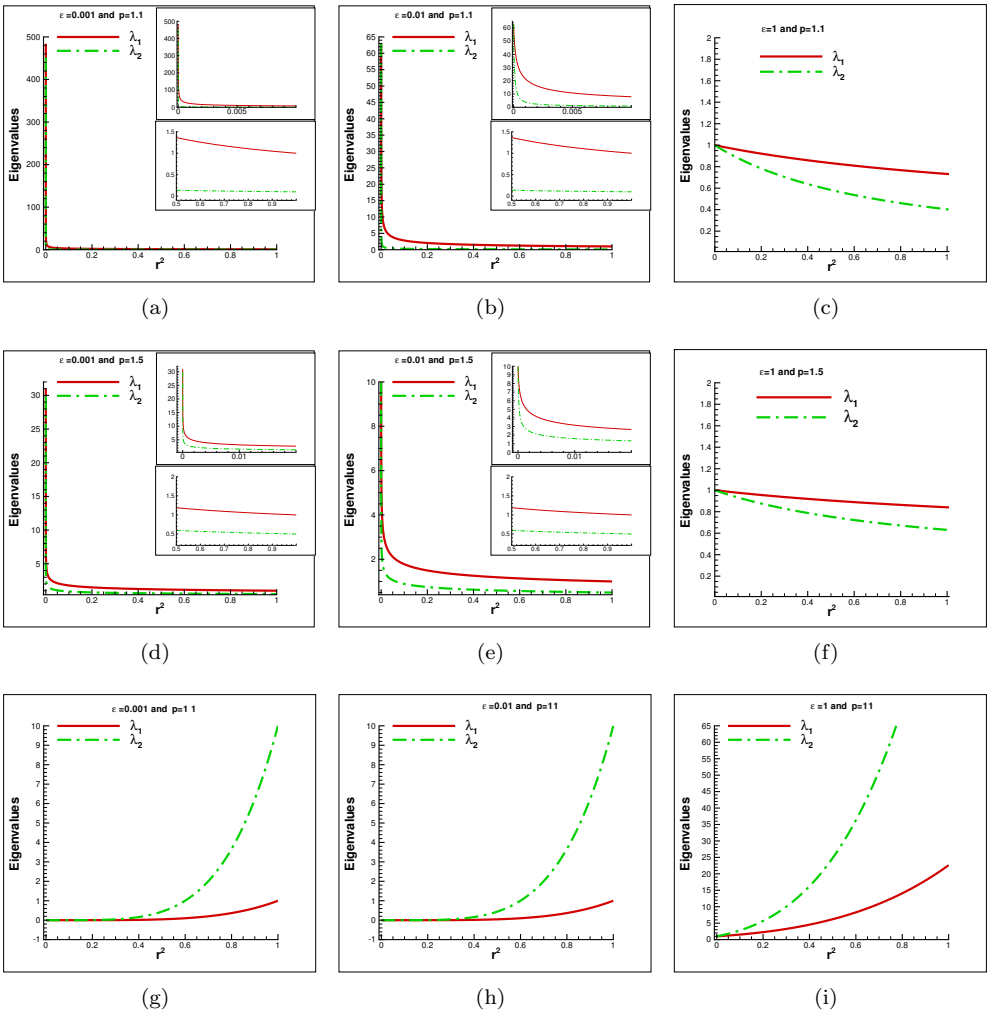


FIG. 1. *Eigenvalues* λ_1, λ_2 *of* $J_{\mathbf{A}}$*:* (a) *plots for* $\varepsilon = 0.001$ *and* $p = 1.1$, (b) *plots for* $\varepsilon = 0.01$ *and* $p = 1.1$, (c) *plots for* $\varepsilon = 1$ *and* $p = 1.1$, (d) *plots for* $\varepsilon = 0.001$ *and* $p = 1.5$, (e) *plots for* $\varepsilon = 0.01$ *and* $p = 1.5$, (f) *plots for* $\varepsilon = 1$ *and* $p = 1.5$, (g) *plots for* $\varepsilon = 0.001$ *and* $p = 11$, (h) *plots for* $\varepsilon = 0.01$ *and* $p = 11$, (i) *plots for* $\varepsilon = 1$ *and* $p = 11$.

nonlinear algebraic problem: Find the vector $\mathbf{U} = [U_1, \ldots, U_i, \ldots, U_{N_h}]$ which satisfies the system of N_h nonlinear equations

$$\mathbf{B}(\mathbf{U}) = \mathbf{f}, \tag{3.4}$$

where the entries $B_i(\mathbf{U})$ of $\mathbf{B}$ and f_i of $\mathbf{f}$ are specified by (2.5), i.e.,

$$B_i(\mathbf{U}) = B(u_h, \phi_{h,i}) \text{ and } f_i = l_f(\phi_{h,i}). \tag{3.5}$$

The nonlinear system (3.4) is generally large sized and hence it is important to develop efficient iterative methods for its numerical solution [37]. Also, it is important to investigate the influence of the parameters p and ε with respect to the convergence speed of the nonlinear iterative procedure.

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Throughout this section, $\mathbf{U}^*$ will denote the exact solution of (3.4) and $\mathbf{U}^n$ the solution derived at step n of the iterative procedure. Furthermore $\|.\|$ will denote the l^2-norm as well as the induced matrix norm. Recall that the condition number of a matrix A relative to the norm $\|.\|$ is given by $\kappa(A) = \|A\|\,\|A^{-1}\|$. We denote the set of eigenvalues of A by $\sigma(A)$ and $\rho(A) = \max\limits_{\lambda\in\sigma(A)} |\lambda|$ is the spectral radius of A.

3.3. Newton-like methods.

3.3.1. Definition and algorithms. In this subsection, we present two Newton techniques for solving (3.4). Let $v_h = \sum\limits_{i=1}^{N_h} V_i\phi_{h,i}$ be an element in $V_{D,h}^{(k)}$ with $\mathbf{V} = [V_1, \ldots, V_i, \ldots, V_{N_h}]$. Using (2.8) we define

$$\mathbf{B}'(\mathbf{V})_{j,i} := B'(v_h)(\phi_{h,i}, \phi_{h,j}),\ i,j = 1,\ldots,N_h, \tag{3.6}$$

to be the iterative matrix of Newton's method. Then the main steps of a Newton method (in a matrix-vector form) can described as follows.

ALGORITHM 3.1 (Newton's method as a defect-correction scheme). *Choose an initial Newton guess* $\mathbf{U}^0$*:*

1. *For* $n = 0, 1, 2, \ldots, n_{max}$ *compute the solution* $\mathbf{Z}^n$ *from*

$$\mathbf{B}'(\mathbf{U}^n)\mathbf{Z}^n = -\big(\mathbf{B}(\mathbf{U}^n) - \mathbf{f}\big), \tag{3.7a}$$

Determine $\lambda_n \in (0, 1]$ (*e.g., see Algorithm* 3.2)

$$\textit{Update:}\quad \mathbf{U}^{n+1} = \mathbf{U}^n + \lambda_n\mathbf{Z}^n. \tag{3.7b}$$

2. *If the stopping criterion (see either* (3.8) *or* (3.9) *below) is satisfied then exit and set* $\mathbf{U}^* := \mathbf{U}^{n+1}$.
Otherwise repeat 1.

Usually the stopping criterion is related to a control of the magnitude of the residual of the new iteration $\mathbf{U}^{n+1}$, i.e,

$$res_{n+1} := \|\mathbf{B}(\mathbf{U}^{n+1}) - \mathbf{f}\| \le TOL, \tag{3.8}$$

where $TOL \sim 10^{-12}$. Based on that we determine the parameter λ_n for the updating step (3.7b). However, in the literature (see, e.g., [29, 37]), more sophisticated Newton methods have been proposed, where the stopping criterion, the damping parameter, and the updating step are determined by estimating the magnitude of the norms

$$\|\mathbf{Z}^n\| \le TOL \quad \text{or} \quad \|\mathbf{Z}_{simp}^{n+1}\| \le TOL, \tag{3.9}$$

where $\mathbf{Z}_{simp}^{n+1}$ is a Newton update from solving a simplified problem; see (3.12) below.

Thus, we formulate two different Newton procedures: (i) a residual-based line-search procedure, which is related to the first case (3.8); see Algorithm 3.2; and (ii) an error-oriented Newton procedure, which is related to the second case (3.9); see Algorithm 3.3.

ALGORITHM 3.2 (residual-based line-search). *In this procedure,* λ_n *is specified by a direct decreasing of the residual in each n-step:*

1. *Set* $\lambda_{n,l=0} := \lambda_{initial} = 1$.
2. *For* $l = 0, \ldots, l_M$, *compute* $\mathbf{Z}^n$ *by* (3.7a) *and update* $\mathbf{U}^{n+1,l} = \mathbf{U}^n + \lambda_{n,l}\mathbf{Z}^n$.
3. *Evaluate* $res_{n,l+1}$.

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4. *If* $res_{n,l+1} \le res_{n,l}$ *then solution found* $\mathbf{U}^{n+1} := \mathbf{U}^{n+1,l}$, *otherwise go to step* 2, *setting* $\lambda_{n,l+1} = \frac{\lambda_{n,l}}{2}$.
5. *If* $l + 1 > l_M$ *stop. Convergence failure.*

ALGORITHM 3.3 (error-oriented Newton's method). *This procedure is based on a natural monotonicity test, i.e.,* $\|\mathbf{Z}^{n+1}_{simp}\| < \|\mathbf{Z}^n\|$.

1. *Choose an initial guess* $\mathbf{U}^0$. *Set the minimal damping factor, e.g.,* $\lambda_{min} \sim 10^{-8}$ *and* $\lambda_0 < 1$. *For* $n = 0, 1, 2, 3, \ldots$:
2. *Solve*
$$\mathbf{B}'(\mathbf{U}^n)\mathbf{Z}^n = -\big(\mathbf{B}(\mathbf{U}^n) - \mathbf{f}\big) \quad (\textit{see Newton step } (3.7\text{a}) \ ).$$
If $\|\mathbf{Z}^n\| \le TOL$ *then the solution is found and set*
$$\mathbf{U}^* := \mathbf{U}^n + \mathbf{Z}^n.$$
3. *If* $\|\mathbf{Z}^n\| > TOL$, *then take* $\mathbf{Z}^n_{simp}$, *which has been already computed in the previous Newton step* $n - 1$ *via* (3.12). *Determine a new prediction value* λ_n *for the damping factor as follows:*
$$\lambda_n := \min(1, \mu_n), \quad \textit{where} \quad \mu_n := \frac{\|\mathbf{Z}^{n-1}\| \cdot \|\mathbf{Z}^n_{simp}\|}{\|\mathbf{Z}^n_{simp} - \mathbf{Z}^n\| \cdot \|\mathbf{Z}^n\|}.$$
If
$$\lambda_n < \lambda_{min} \tag{3.10}$$
then stop. Convergence failure and abort computation.
4. *If* $\lambda_n > \lambda_{min}$ *continue and compute trial iterate*
$$\mathbf{U}^{n+1} := \mathbf{U}^n + \lambda_n \mathbf{Z}^n \tag{3.11}$$
and evaluate the new residual $-(\mathbf{B}(\mathbf{U}^{n+1}) - \mathbf{f})$. *Solve the simplified linear system using the old Jacobian and the new residual*
$$\mathbf{B}'(\mathbf{U}^n)\mathbf{Z}^{n+1}_{simp} = -\big(\mathbf{B}(\mathbf{U}^{n+1}) - \mathbf{f}\big). \tag{3.12}$$
5. *Compute the monitoring functions:*
$$\theta_n := \frac{\|\mathbf{Z}^{n+1}_{simp}\|}{\|\mathbf{Z}^n\|}, \quad \mu'_n := \frac{0.5\|\mathbf{Z}^n\|\,\lambda_n^2}{\|\mathbf{Z}^{n+1}_{simp} - (1 - \lambda_n)\mathbf{Z}^n\|}.$$
6. *If* $\theta_n \ge 1$ *(no convergence of the updates), then set*
$$\lambda_n := \lambda'_n := \min\left(\mu'_n, \frac{1}{2}\lambda_n\right)$$
and go to (3.10) *and continue from there.*
7. *Else, if* $\theta_n < 1$, *we have convergence and continue with the next steps.*
Set $\lambda'_n := \min(1, \mu'_n)$.
   (a) *If* $\lambda'_n = \lambda_n = 1$ *then: if* $\|\mathbf{Z}^{n+1}_{simp}\| \le TOL$ *stop and the solution is found:*
$$\mathbf{U}^* := \mathbf{U}^{n+1} + \mathbf{Z}^{n+1}_{simp},$$
otherwise go to the beginning of step 2 *and increment* $n \to n + 1$.
   (b) *If* $\lambda'_n \ne \lambda_n < 1$ *go to the beginning of step* 2 *and increment* $n \to n + 1$.

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Remark 3.2 (motivation of steps 3 and 5). In the above algorithm, the important steps are 3 and 5. Both are based on adaptive trust region strategies as derived in [29, section 3.3.3]. Specifically, step 3 is a prediction strategy that is based on an estimate of the simplified update and the full Newton update, while using Lipschitz continuity of the Jacobian. Step 5 is a correction strategy that is based on a damped Newton iteration.

We underline that for finding the numerical solution $\mathbf{U}^*$ the generated global error includes two parts, one coming form the finite element discretization and the second coming by the Newton iterative procedure. Recalling (2.33) and the stopping criterion in the Newton algorithms, we can practically express the global error as

$$\|\mathbf{F}(\nabla u) - \mathbf{F}(\nabla u_{h,n})\| \lesssim \mathcal{O}(h^{l-1}) + res_n, \tag{3.13}$$

where res_n is the residual at the last nth Newton iteration step. Thus, the Newton iterations must be continued until the discretization error component dominates.

Remark 3.3. In view of Jacobian $J_{\mathbf{A}}$ defined in (3.2), the entries of the iterative matrix in (3.6) are equal to $\mathbf{B}'(\mathbf{V})_{j,i} = \int_\Omega J_{\mathbf{A}}(\nabla v_h)\nabla\phi_{h,i}\nabla\phi_{h,j}\,dx$ and, consequently, Newton's step in (3.7a) can be expressed as find $z_h \in V_{D,h}^{(k)}$ such that

$$\begin{aligned}\int_\Omega J_{\mathbf{A}}(\nabla u_h^n)\nabla z_h \cdot \nabla\phi_h\,dx \\ = -\bigg(\int_\Omega \mathbf{A}(\nabla u_h^n)\cdot\nabla\phi_h\,dx - \int_\Omega f\phi_h\,dx\bigg), \quad \phi_h \in V_{0,h}^{(k)},\end{aligned} \tag{3.14}$$

where $u_h^n \in V_{D,h}^{(k)}$ is the updated numerical solution of the previous step.

3.3.2. Remarks on the properties for the condition number of $\mathbf{B}'$. In practice we can check the well-conditioned status of system (3.4) (and also (3.7a)) through the condition number. The condition number $\kappa(\mathbf{B}')$ is a measure of sensitivity of the solution to relatively small perturbations of the system, e.g., finite precision arithmetic, as well as a measure of the distance from a singular matrix. Roughly speaking, an almost singular matrix is expected to have a large condition number and thus we expect to meet problems during the solution of the system. In our studies this could be the case where $p > 2$ and $\varepsilon \to 0$; see Figures 1(g),(h). Next, we quote a Lemma related to inverse estimates for $\phi_h \in V_{D,h}^{(k)}$.

LEMMA 3.4. *Let $\phi_h = \sum_{i=1}^{N_h} v_i\phi_{h,i}$ and $\mathbf{v} = (v_1, \ldots, v_{N_h})$. There exist positive constants C_0, C_m, and C_M depending only on the quasi-uniform properties of the meshes and on k, such that*

$$\|\nabla\phi_h\|_{L^2(\Omega)} \le C_0 h^{-1}\|\phi_h\|_{L^2(\Omega)}, \tag{3.15a}$$

$$C_m h\|\mathbf{v}\| \le \|\phi_h\|_{L^2(\Omega)} \le C_M h\|\mathbf{v}\|, \tag{3.15b}$$

where $\|\mathbf{v}\|^2 = v_1^2 + \cdots + v_{N_h}^2$.

Proof. Both relations can be shown by applying scaling arguments; see details in [28]. ∎

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Based on (2.8), we can clearly see that the matrix $\mathbf{B}'$ defined in (3.6) is symmetric and is positive definite. Indeed by (3.6) and (2.8), we have

$$B'(u_h)(w_h,w_h) = \int_\Omega (\varepsilon^2+|\nabla u_h|^2)^{\frac{p-2}{2}}|\nabla w_h|^2\,dx + (p-2)\int_\Omega (\varepsilon^2+|\nabla u_h|^2)^{\frac{p-4}{2}}(\nabla u_h,\nabla w_h)^2\,dx. \tag{3.16}$$

Now, let us try to derive bounds for $\frac{B'(u_h)(w_h,w_h)}{\|\mathbf{w}\|^2}$. As a result, using that $(a^2+\beta^2)\le (a+\beta)^2\le 2(a^2+\beta^2)$ for $a,\,\beta>0$, we have that

$$\begin{aligned} B'(u_h)(w_h,w_h) &\le \int_\Omega (\varepsilon^2+|\nabla u_h|^2)^{\frac{p-2}{2}}|\nabla w_h|^2\,dx \\ &\quad + |p-2|\int_\Omega (\varepsilon^2+|\nabla u_h|^2)^{\frac{p-4}{2}}(\varepsilon^2+|\nabla u_h|^2)|\nabla w_h|^2\,dx \\ &\le C(p)\int_\Omega (\varepsilon^2+|\nabla u_h|^2)^{\frac{p-2}{2}}|\nabla w_h|^2\,dx. \end{aligned} \tag{3.17}$$

If $p>2$ we get immediately that

$$B'(u_h)(w_h,w_h) \ge \int_\Omega (\varepsilon^2+|\nabla u_h|^2)^{\frac{p-2}{2}}|\nabla w_h|^2\,dx. \tag{3.18}$$

For the case $1<p<2$, a simple modification of the last term in (3.16) yields

$$\begin{aligned} B'(u_h)(w_h,w_h) &\ge \int_\Omega (\varepsilon^2+|\nabla u_h|^2)^{\frac{p-2}{2}}|\nabla w_h|^2\,dx \\ &\quad + (p-2)\int_\Omega (\varepsilon^2+|\nabla u_h|^2)^{\frac{p-4}{2}}|\nabla u_h|^2\,|\nabla w_h|^2\,dx \\ &\ge (p-1)\int_\Omega (\varepsilon^2+|\nabla u_h|^2)^{\frac{p-2}{2}}|\nabla w_h|^2\,dx \\ &> 0. \end{aligned} \tag{3.19}$$

In the following, inspired by [31], we try to derive estimates for the condition number $\kappa(\mathbf{B}')$.

PROPOSITION 3.5. *The condition number $\kappa(\mathbf{B}')$ can be estimated by*

$$\kappa(\mathbf{B}') = \frac{\lambda_{Max}}{\lambda_{min}} \le \begin{cases} C(\varepsilon+|\nabla u_h|_\infty)^{p-2}\,\varepsilon^{2-p}\,h^{-2} & \text{for } p>2,\\ C(\varepsilon+|\nabla u_h|_\infty)^{2-p}\,\varepsilon^{p-2}\,h^{-2} & \text{for } 1<p<2, \end{cases} \tag{3.20}$$

where λ_{Max} and λ_{min} are the largest and the smallest eigenvalue of $\mathbf{B}'$, respectively.

Proof. For $p>2$, it follows from (3.17) and (3.15) that

$$\begin{aligned} \frac{B'(u_h)(w_h,w_h)}{\|\mathbf{w}\|^2} &\le C(\varepsilon+\|\nabla u_h\|_\infty)^{p-2}\frac{|\nabla w_h|^2_{L^2(\Omega)}}{h^{-2}\|w_h\|^2_{L^2(\Omega)}} \\ &\le C(\varepsilon+\|\nabla u_h\|_\infty)^{p-2}\frac{h^{-2}\|w_h\|^2_{L^2(\Omega)}}{h^{-2}\|w_h\|^2_{L^2(\Omega)}} \\ &\le C(\varepsilon+|\nabla u_h|_\infty)^{p-2}. \end{aligned} \tag{3.21}$$

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For $p > 2$ using (3.18) and (2.9b), we obtain

$$\frac{B'(u_h)(w_h, w_h)}{\|\mathbf{w}\|^2} \geq C\varepsilon^{p-2} \frac{|\nabla w_h|^2_{L^2(\Omega)}}{h^{-2}\|w_h\|^2_{L^2(\Omega)}} \geq C\varepsilon^{p-2} \frac{\|w_h\|^2_{L^2(\Omega)}}{h^{-2}\|w_h\|^2_{L^2(\Omega)}} \geq C\varepsilon^{p-2}h^2. \tag{3.22}$$

On the other hand, for $1 < p < 2$ we have

$$\frac{B'(u_h)(w_h, w_h)}{\|\mathbf{w}\|^2} \leq C\varepsilon^{p-2} \frac{|\nabla w_h|^2_{L^2(\Omega)}}{h^{-2}\|w_h\|^2_{L^2(\Omega)}} \leq C\varepsilon^{p-2} \frac{h^{-2}\|w_h\|^2_{L^2(\Omega)}}{h^{-2}\|w_h\|^2_{L^2(\Omega)}} \leq C\varepsilon^{p-2}. \tag{3.23}$$

Also, for $1 < p < 2$ using (3.19) and (2.9b), we obtain

$$\begin{aligned} \frac{B'(u_h)(w_h, w_h)}{\|\mathbf{w}\|^2} &\geq C(\varepsilon + \|\nabla u_h\|_\infty)^{p-2} \frac{|\nabla w_h|^2_{L^2(\Omega)}}{h^{-2}\|w_h\|^2_{L^2(\Omega)}} \\ &\geq C(\varepsilon + \|\nabla u_h\|_\infty)^{p-2} \frac{\|w_h\|^2_{L^2(\Omega)}}{h^{-2}\|w_h\|^2_{L^2(\Omega)}} \\ &\geq C(\varepsilon + |\nabla u_h|_\infty)^{p-2}h^2. \end{aligned} \tag{3.24}$$

Gathering together all the previous inequalities we can show (3.20). ∎

Remark 3.6. We note that a necessary condition for obtaining the solution $\mathbf{Z}^n$ during the Newton process is that the iterative matrix $\mathbf{B}'$ is invertible; see Algorithm 3.1. In the proofs of the convergence of Newton's methods that have been presented in the literature, the existence of $\mathbf{B}'^{-1}$ (at least in a neighborhood of $\mathbf{U}^*$) is usually supposed. Additional assumptions are the existence of $\beta_m > 0$ and $\beta_M > 0$ such that $\beta_m \leq \|\mathbf{B}'(\mathbf{V})^{-1}\| \leq \beta_M$. Usually, the bound β_M is used to define the radius of the area around $\mathbf{U}^*$ for choosing the initial $\mathbf{U}^0$; in order to be able to show analytically the convergence of the method, see [29, 37]. From (3.20) and (3.3), one sees that for our case it is difficult to find uniform bounds with respect to ε and p. In the numerical examples, we observed that, indeed, the choice of $\mathbf{U}^0$ influences the performance of the Newton solver. Choosing the initial guess to be the projection of the previous mesh solution, we have a fast convergence of the iterative procedure; see, for instance, Table 2 of Example 1.

3.4. Augmented Lagrangian techniques. In the following two paragraphs, we transform the original problem (2.1) into a saddle point problem using ALG techniques. Let q be the conjugate exponent of p, that is $\frac{1}{p} + \frac{1}{q} = 1$, and let us define the space $W \subset W^{1,p}_D \times (L^p(\Omega))^2$ by

$$W = \{(v, \mathbf{q}) | (v, \mathbf{q}) \in W^{1,p}_D \times (L^p(\Omega))^2 : \nabla v - \mathbf{q} = 0\}. \tag{3.25}$$

Following [38], we introduce the ALG $\mathcal{L}_{\widehat{r}}$ defined, for $\widehat{r} > 0$, by

$$\begin{aligned} \mathcal{L}_{\widehat{r}}(v, \mathbf{q}, \lambda) = \frac{1}{p}\int_\Omega (\varepsilon^2 + |\mathbf{q}|^2)^{\frac{p}{2}}\, dx - \int_\Omega fv\, dx \\ + \frac{\widehat{r}}{2}\int_\Omega |\nabla v - \mathbf{q}|^2\, dx + \int_\Omega \lambda \cdot (\nabla v - \mathbf{q})\, dx, \end{aligned} \tag{3.26}$$

and the following associated with saddle point problem (3.26):

$$\text{Find} \{u, \mathbf{q}, \lambda\} \in W^{1,p}_D \times (L^p(\Omega))^2 \times (L^q(\Omega))^2 \text{ such that} \tag{3.27a}$$

$$\mathcal{L}_{\widehat{r}}(u, \mathbf{q}, \mu) \leq \mathcal{L}_{\widehat{r}}(u, \mathbf{q}, \lambda) \leq \mathcal{L}_{\widehat{r}}(v, \mathbf{w}, \lambda), \tag{3.27b}$$

$$\forall \{v, \mathbf{w}, \mu\} \in W^{1,p}_D \times (L^p(\Omega))^2 \times (L^q(\Omega))^2. \tag{3.27c}$$

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TABLE 1
The steps of the two ALG1 algorithms.

Algorithm sALG1:	**Algorithm mALG1:**
for $\lambda^0 \in (L^q(\Omega))^2$ given **For** each ALG iterative step $i > 0, \in \mathbb{N}$ find $u^i := u^{i,n_M}$, $\mathbf{q}^i := \mathbf{q}^{i,n_M}$ where n_M is the final index n for which the inner Newton loop is converged. **For** each Newton step $n = 0, 1, 2, \ldots$ solve for fixed i: $\begin{cases} -\widehat{r}\Delta u^{i,n} + \widehat{r}\nabla \cdot \mathbf{q}^{i,n} = \nabla \cdot \lambda^i + f, \\ (\varepsilon^2 + \lvert\mathbf{q}^{i,n}\rvert^2)^{\frac{p-2}{2}}\mathbf{q}^{i,n} + \widehat{r}\mathbf{q}^{i,n} - \widehat{r}\nabla u^{i,n} = \lambda^i, \end{cases}$ **EndFor Newton loop** **update**: $\lambda^{i+1} = \lambda^i + \widehat{\rho}(\nabla u^{i,n_M} - \mathbf{q}^{i,n_M})$ **EndFor ALG loop**	for $\lambda^0 \in (L^q(\Omega))^2$ given **For** each iterative Newton step $n > 0 \in \mathbb{N}$ find u^n, $\mathbf{q}^n$ and λ^n as follows $\begin{cases} -\widehat{r}\Delta u^n - \nabla \cdot \lambda^n + \widehat{r}\nabla \cdot \mathbf{q}^n = f, \\ ((\varepsilon^2 + \lvert\mathbf{q}^n\rvert^2)^{\frac{p-2}{2}}\mathbf{q}^n + \widehat{r}\mathbf{q}^n - \widehat{r}\nabla u^n - \lambda^n = 0, \\ \lambda^n - \widehat{\rho}(\nabla u^n - \mathbf{q}^n) = \lambda^{n-1}, \end{cases}$ **EndFor**

In that way, we reduce the solution of (2.1) to finding the saddle point solutions of $\mathcal{L}_{\widehat{r}}$. We apply a variational analysis to (3.27) and then for computing the saddle point solutions, we employ two variants of the first ALG iterative algorithm, (ALG1), described in [32]. The two algorithms are listed in Table 1, where $0 < \widehat{\rho} < 2\,\widehat{r}$. In the first case the Lagrangian λ is updated separately by applying a postprocessing procedure at the end of every iterative circle, which has been previously performed for computing $u, \mathbf{q}$. We call the first version as splitting ALG1 (sALG1), and we present the algorithm in the left column of Table 1. In the second iterative ALG1 procedure, all the unknowns are simultaneously computed at every step, and we call this procedure monolithic ALG1 (mALG1). The algorithm of the mALG1 iterative method is given in the right column of Table 1. The analysis for showing the existence and uniqueness of the saddle point problems seems to be quite hard, as well as the study of the convergence properties of the resulting ALG methods. Such results are discussed in [32] for several problems set in $W^{2,2}$.

At first glance the ALG1 technique seems not to be so attractive due to the introduction of the two variables $\mathbf{q}$ and λ. But as we can see from Table 1, the ALG1 method helps us to simplify the nonlinear structure of the original problem. In particular we are able to uncouple the strong nonlinearity of the model and the derivatives of u. The convergence properties of Algorithm sALG1 have been studied in [32] for $\widehat{r} > 0$. Recalling the finite dimensional spaces defined in (2.13), the discrete variational form of the sALG1 method is as follows: For λ_h^0 given, find $(u_h^i, \mathbf{q}_h^i)$ such that

$$\widehat{r}(\nabla u_h^{i,n}, \nabla v_h) + (\widehat{r}\nabla \cdot \mathbf{q}_h^{i,n}, v_h) = (\nabla \cdot \lambda_h^i + f, v_h) \quad \forall v_h \in V_{0,h}^{(k)}, \tag{3.28a}$$

$$((\varepsilon^2 + \lvert\mathbf{q}_h^{i,n}\rvert^2)^{\frac{p-2}{2}}\mathbf{q}_h^{i,n}, \phi_h) - \widehat{r}(\nabla u_h^{i,n} - \mathbf{q}_h^{i,n}, \phi_h) = (\lambda_h^i, \phi_h) \quad \forall \phi_h \in (V_h^{(k)})^2, \tag{3.28b}$$

$$\text{Update: } \lambda_h^{i+1} = \lambda_h^i + \widehat{\rho}(\nabla u_h^{i,n_M} - \mathbf{q}_h^{i,n_M}). \tag{3.28c}$$

Following a standard finite element methodology, as in section 3.2, we obtain the following nonlinear algebraic system

$$\mathbf{B}_{sALG1}\begin{pmatrix}\mathbf{U}\\ \mathbf{Q}\end{pmatrix} = \mathbf{f}_{sALG1}, \tag{3.29}$$

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where $\mathbf{U}$ and $\mathbf{Q}$ are the vectors with the degrees of freedom of $u_h^{i,n}$ and $\mathbf{q}_h^{i,n}$ respectively. A similar discrete variational form can be derived for the mALG1 method and then we get the following nonlinear algebraic system

$$\mathbf{B}_{mALG1}\begin{pmatrix}\mathbf{U}\\ \mathbf{Q}\\ \mathbf{\Lambda}\end{pmatrix} = \mathbf{f}_{mALG1}, \tag{3.30}$$

where $\mathbf{U}$, $\mathbf{Q}$, and $\mathbf{\Lambda}$ are the vectors with the degrees of freedom of u_h^n and $\mathbf{q}_h^n$ and λ_h^n, respectively.

Remark 3.7. The resulting nonlinear algebraic systems (3.29) and (3.30) can be solved by utilizing any of the two Newton variants; see Algorithm 3.2 and Algorithm 3.3. In the numerical examples, we use Algorithm 3.2 where the stopping criterion is defined in an analogous way to (3.8). In particular, the stopping criterion for the ALG iterative loop in sALG1 is defined as

$$\max\{\|\mathbf{U}^{i+1}-\mathbf{U}^i\|, \|\mathbf{Q}^{i+1}-\mathbf{Q}^i\|, \|\mathbf{\Lambda}^{i+1}-\mathbf{\Lambda}^i\|\} \leq 10^{-6}.$$

Remark 3.8. We notice that all linear equation systems in this paper are solved with the direct solver UMPACK [39]. We are currently working on developing iterative solvers based on geometric multigrid.

Remark 3.9. In the numerical examples, the parameters $\widehat{r}$ and $\widehat{\rho}$ have been specified after carrying out a few computations using different values. As outlined in [32], the optimal choice for $\widehat{r}$ is an open question. Through the performance of the numerical tests, we have observed that the method sALG1 is more sensitive to the choice of $\widehat{r}$ than mALG1. The mALG1 method appears to have greater robustness on the choice of $\widehat{r}$ and $\widehat{\rho}$ with respect to p. In the numerical examples, we have used $\widehat{\rho}=\widehat{r}=1$ for sALG1 and $\widehat{r}=100$, $\widehat{\rho}=0.1$ for mALG1. Choosing this high value for $\widehat{r}$, in fact, we increase the strength of the penalization in the mALG1 method.

4. Numerical examples. In this section, we perform several numerical tests by selecting different values for the exponent p and investigate the order of accuracy of the proposed finite element method. We mainly focus on cases where $1<p<2$ but do also consider a few tests with $p>2$. The examples have been performed using first ($k=1$) and second ($k=2$) order polynomial spaces. We compare the error convergence rates versus the grid size for several values of ε ranging between 10^{-4} and 1. Each example has been solve applying several mesh refinement steps with $h_i, h_{i+1}, \ldots$, satisfying Assumption 2.6. The numerical convergence rates r have been computed by the ratio $r=\frac{ln(e_i/e_{i+1})}{ln(h_i/h_{i+1})}$, $i=1,2,\ldots$, where the corresponding error is $e_i := \|\mathbf{F}(\nabla u)-\mathbf{F}(\nabla u_{h,n})\|_{L^2}$ (see (2.33)), and $u_{h,n}$ is the final solution computed by the iterative procedure. We mention that, in the test cases with smooth solutions, i.e., $k+1\leq l$, the approximation order in (2.33) is expected to be equal to k. The columns in the tables below which are related to N indicate the maximum number of nonlinear iterative steps. All tests have been performed with the C++ library deal.II [35, 36] using an Intel(R) Core(TM) i5-3320M CPU 2.60 GHz computer.

4.1. Newton-like methods.

Example 1 *(*$|\nabla u| \neq 0 \,\forall\, x \in \Omega$*).* The first numerical example is a simple test case for validating the error estimates given in Theorem 2.12 and demonstrating the performance of the Newton method for several values of ε. We take $\Omega \subset \mathbb{R}^2$ to be the square domain $(0, \frac{\pi}{2})^2$. The boundary data u_D and the function f in (2.1), are

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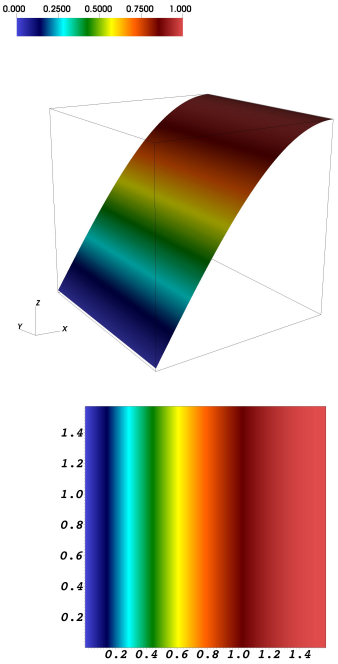


FIG. 2. *Example* 1. *Display of the numerical solution for the case* $p = 1.01$ *and* $\varepsilon = 1.0e-4$.

chosen so that the exact solution in (2.1) will be $u(x, y) = \sin(x)$. The exact solution is shown in Figure 2 and it holds that $\{x \in \Omega : \nabla u = 0\} = \emptyset$.

We compute numerically the convergence rates of the error on a sequence of successively refined meshes and we also investigate the performance of the residual-based Newton method: for different choices of ε. Here we consider two different choices for the starting solution u_h^0 for Newton methods: (i) we set $u_h^0 = 0$; (ii) $u_h^0 = Proj(u_{2h}^n)$, where $Proj(u_{2h}^n)$ is the projection of the solution u_{2h}^n, which has been computed on the previous space $V_{D,2h}^{(k)}$, to the current space $V_{D,h}^{(k)}$. Here the index n denotes the last (i.e., converged) solution on the coarser space $V_{D,2h}^{(k)}$. The projection is carried out as nodal interpolation from the coarse mesh on the finer mesh.

In Table 2, we display the results for polynomial degree $k = 1$ for $p = 1.01$ and for $\varepsilon = 10^{-3}$ (right columns) and $\varepsilon = 10^{-4}$ (left columns). For both ε cases, we observe that the rates r related to the first coarse meshes are little less than the expected rates, but progressively as the mesh is refined, the rates tend to the optimal value $r = 1$, which is in agreement with Theorem 2.12. For the case $u_h^0 = 0$, we observe that the Newton method is not mesh independent in the sense that the number of steps is not stable as the mesh is further refined. On the other hand, for the case $u_h^0 = Proj(u_{2h}^n)$, we observe mesh independence and the further number of steps is significantly reduced; see also comments in Remark 3.6. For theoretical results on mesh independence for general nonlinear problems we refer the reader to [40]. As expected, this behavior of the Newton method affects the corresponding CPU times, which are provided in the last row of Table 2. If not further indicated in all remaining tests, we always use as initial guess $u_h^0 = Proj(u_{2h}^n)$.

We repeat the previous computations choosing $\varepsilon = 10^{-2}$ and $\varepsilon = 1$. The results are shown in Table 3. We can clearly see that the rates r are in full agreement

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TABLE 2
Example 1. *The results for the different values of* ε.

-	$k=1$ and $p=1.01$							
-	$\varepsilon = 10^{-4}$				$\varepsilon = 10^{-3}$			
DoFs	$\|F - F_h\|$	r	$N, u_h^0 = 0$	N, $u_h^0 = P(u_{2h}^n)$	$\|F - F_h\|$	r	$N, u_h^0 = 0$	N, $u_h^0 = P(u_{2h}^n)$
289	3.64207e-02	-	16	16	3.66426e-02	-	13	13
1089	1.96867e-02	0.88	18	7	2.15775e-02	0.76	16	8
4225	1.05728e-02	0.9	22	8	1.28714e-02	0.75	23	9
16641	5.69064e-03	0.9	26	7	6.77354e-03	0.93	28	9
66049	3.17429e-03	0.85	32	8	3.31612e-03	1.03	32	8
263169	1.78568e-03	0.83	43	10	1.65509e-03	1.0	34	7
1050625	9.36320e-04	0.94	47	8	8.54432e-04	0.96	36	6
CPU	-	-	1.33e+04s	2.13e+03s	-	-	9.87e+03s	1.80e+03s

TABLE 3
Example 1. *The results for the different values of* ε.

-	$k=1$ and $p=1.01$					
-	$\varepsilon = 10^{-2}$			$\varepsilon = 1$		
DoFs	$\|F - F_h\|$	r	N	$\|F - F_h\|$	r	N
289	4.64339e-02	-	12	2.77038e-02	-	12
1089	2.27756e-02	1.02	8	1.38533e-02	0.99	5
4225	1.16848e-02	0.96	9	6.92684e-03	1.0	5
16641	6.05204e-03	0.95	10	3.46344e-03	1.0	3
66049	3.04030e-03	0.99	8	1.73172e-03	1.0	3
263169	1.51975e-03	1.0	5	8.65862e-04	1.0	3
1050625	7.59810e-04	1.0	4	4.32931e-04	1.0	2
CPU	-	-	8.46e+02s	-	-	5.07e+02s

with Theorem 2.12. The Newton method is mesh independent and, in particular, for the case $\varepsilon = 1$ only a few steps (less than 3) are needed. In comparison to the corresponding data in Table 2, the CPU times are reduced in Table 3 as a result of fewer iterative steps. By the results of the two previous numerical tests, we conclude that the rates r are very close to the expected value for all ε but the Newton steps are increased as the values of ε become smaller.

Next, we investigate the convergence rate of the error and the performance of the residual-based Newton method for $p = 1.1$ and for three different choices of ε. The results are shown in Table 4. Here, we can see that for each of the three values of ε, the error $\|\mathbf{F}(\nabla u) - \mathbf{F}(\nabla u_h^n)\|_{L^2}$ converges to zero with rate r close to one, which is in agreement with Theorem 2.12. The Newton performance appears to be mesh independent. The required total steps are increased as the value of ε is decreased.

As a last computation in this example, we solve the problem using $k = 2$ polynomials. The parameter p takes the limiting values $p = 1.01$ and $p = 1.1$. Again, as in the above tests, we investigate the behavior of the convergence of the error as well the behavior of the iterative solver for several values of ε. The results are displayed in Table 5 and in Table 6. Here, we observe for $\varepsilon = 10^{-4}$ that the error converges with the expected rate r for both p-cases. We repeat the computations setting $\varepsilon = 0.1$. Also, in this case the convergence rates of the error tend to obtain the optimal value $r = 2$ as the mesh is refined. For both p-cases, the rates are optimal for all mesh refinement levels and the iterative solver needs less than two steps for

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TABLE 4
Example 1. *The results for the different values of* ε.

-	$k=1$ and $p=1.1$								
-	$\varepsilon=10^{-4}$			$\varepsilon=10^{-3}$			$\varepsilon=10^{-1}$		
DoFs	$\|F-F_h\|$	r	N	$\|F-F_h\|$	r	N	$\|F-F_h\|$	r	N
289	3.61674e-02	-	11	3.62331e-02	-	10	3.90390e-02	-	9
1089	1.92142e-02	0.91	5	1.93753e-02	0.9	5	1.95622e-02	1.0	4
4225	1.00983e-02	0.93	5	1.04294e-02	0.9	6	9.78098e-03	1.0	4
16641	5.26816e-03	0.94	5	5.64053e-03	0.88	5	4.89046e-03	1.0	3
66049	2.73798e-03	0.95	5	2.89079e-03	0.96	6	2.44523e-03	1.0	3
263169	1.42804e-03	0.94	6	1.46773e-03	0.98	5	1.22261e-03	1.0	3
1050625	7.44909e-04	0.94	8	7.51022e-04	0.97	5	6.11307e-04	1.0	2
CPU	-	-	1.05e+03s	-	-	7.27e+02s	-	-	3.47e+02s

TABLE 5
Example 1. *The results for* $k=2$ *for different choices of* ε.

-	$k=2$ and $p=1.01$					
-	$\varepsilon=10^{-4}$ $p=1.01$			$\varepsilon=10^{-1}$ $p=1.01$		
DoFs	$\|F-F_h\|$	r	N	$\|F-F_h\|$	r	N
256	2.221e-04	-	20	2.316e-04	-	11
1024	5.509e-05	2.01	4	5.787e-05	2.00	2
4096	1.382e-05	1.99	4	1.446e-05	2.00	2
16384	3.457e-06	2.00	5	3.616e-06	2.00	1
65536	8.645e-07	2.00	5	9.041e-07	2.00	1
262144	2.161e-07	2.00	5	2.260e-07	2.00	1
CPU	-	-	6.999e+02s	-	-	1.658e+02s

TABLE 6
Example 1. *The results for* $k=2$ *for different choices of* ε.

$k=2$ and $p=1.1$					
$\varepsilon=10^{-4}$ $p=1.1$			$\varepsilon=10^{-1}$ $p=1.1$		
$\|F-F_h\|$	r	N	$\|F-F_h\|$	r	N
2.394e-04	-	14	2.476e-04	-	14
5.939e-05	2.01	4	6.190e-05	2.00	2
1.490e-05	1.99	4	1.547e-05	2.00	2
3.726e-06	2.00	4	3.868e-06	2.00	1
9.318e-07	2.00	4	9.672e-07	2.00	1
2.329e-07	2.00	4	2.418e-07	2.00	1
CPU -	-	5.719e+02s	-	-	1.671e+02s

obtaining the solution. Again, this test illustrates the affection of the parameter ε to the performance of the iterative solver.

Example 2 *(existence of* $x_0 \in \Omega : |\nabla u(x_0)| = 0$*).* In the second example, we want to investigate the convergence rates of the error and the corresponding behavior of the residual-based Newton method in the case of the presence of critical points, i.e., $\{x_0 \in \Omega : |\nabla u(x_0)| = 0\}$. In this situation the iterative matrix $\mathbf{B}'$ in (3.7a) tends to become singular or ill-conditioned as $\varepsilon \to 0$. To this end, the domain is $\Omega = (0,1)^2$ and the data u_D and f are selected so that the exact solution will be given by $u(x,y) = \sin(2\pi(x+y))$, which is plotted in Figure 3. Note that the solution contains critical points along the lines $y = \frac{1}{4} - x$ and $y = \frac{3}{4} - x$. We solved the problem using the $k=1$ polynomial space and initial guess $u_h^0 = Proj(u_{2h}^n)$. We compute the convergence rates on a sequence of seven successively finer meshes for several values

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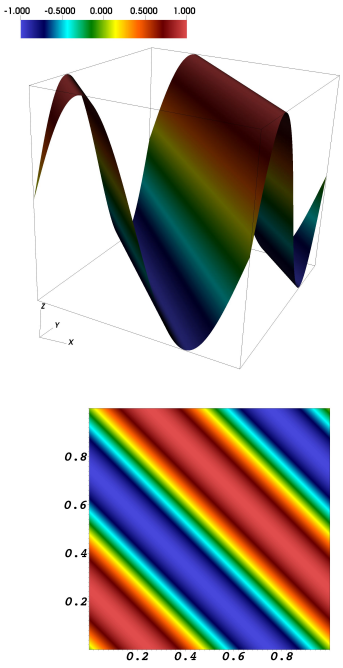


FIG. 3. *Example* 2. *Display of the solution for the case* $p = 1.01$ *and* $\varepsilon = 1.0$.

TABLE 7
Example 2. *The rates* r *and the Newton iteration steps with respect to* ε *for several values of* p.

ε	$p = 1.01$		$p = 1.1$		$p = 1.3$		$p = 1.5$		$p = 1.8$		$p = 2.25$		$p = 3$		$p = 4.3$		$p = 11$	
	r	N	r	N	r	N	r	N	r	N	r	N	r	N	r	N	r	N
10^{-4}							1.3	8	1.08	9	1	6	1	7	1	10	1	20
10^{-3}					1.8	11	1.27	8	1.1	9	1	6	1	7	1	10	1	20
10^{-2}					2.4	12	1.37	8	1.1	6	1	5	1	7	1	9	1	19
10^{-1}			1	7	1	9	1	6	1	5	1	5	1	7	1	6	1	18
1	1	4	1	4	1	5	1	5	1	5	1	5	1	4	1	5	1	9

of p and ε ranging from 10^{-4} to 1. In Table 7, we show the convergence rates r and the Newton iterations for each selected p and each ε. The reported rate r is the minimum rate computed on the last three finer meshes. The reported number of Newton iterations, N, is the maximum number of Newton iterations that have been used on the three finer meshes. We can clearly see that for $p < 2$ the performance of the Newton solver is very sensitive to the choice of ε. In particular, the value of ε must be close to one in order to perform successfully the Newton iterative procedure. An exception is the case of $p = 1.8$ where the problem has been solved for all values of ε using a relatively small number of iterative steps N. On the other hand for the tests with $p > 2$, the problem has been successfully solved for all the selected values of ε. We can see that the rates have the expected values and the Newton method appears to have a mesh independent behavior. For the case $p = 11$, we observe an increase in the Newton steps without having a deterioration of the convergence rates.

Example 3 *(low regularity solution,* $u \in W^{l,p}(\Omega)$, $l = 2$ *and* $l = 3$*).* This example consists of a problem with a low regularity solution, i.e., $u(x, y) = (x^2 + y^2)^{\frac{\gamma}{2}}$ for

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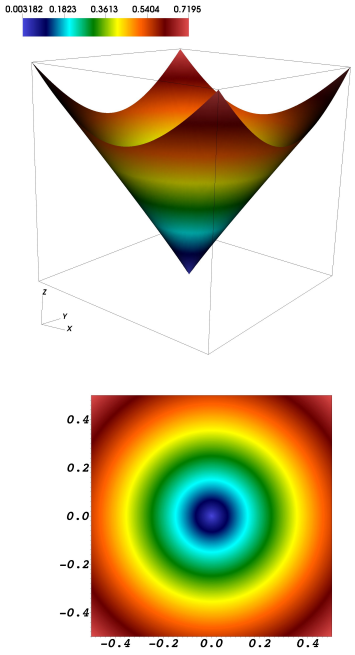


FIG. 4. *Example* 3. *Display of the numerical solution for the case* $p = 1.5$ *and* $\varepsilon = 1.0$ *and* $\gamma = 0.8$ *and* $k = 1$.

several values of the parameter γ; see Figure 4 and [17]. The computational domain is the rectangle $\Omega = (-0.5, 0.5)^2$. The source function f and u_D are manufactured by the exact solution. We point out that the regularity of the solution is specified by the value of the parameter γ and, in this example, we investigate the ability of the whole method (finite element discretization and Newton solver) to approximate low regularity solutions of (2.1) with the expected accuracy. Thus, this example seems to be quite interesting, since we can predefine the regularity of the proposed exact solution to be close to the required regularity of the solution, which has been assumed in the error analysis; see Assumptions 2.1, 2.6, and Remark 2.7. In Table 8, for several values of the parameters γ and p, we display the space $W^{l,p}$ which the solution u belongs to. Also, the expected convergence rates r are mentioned. For validation, we have computed the convergence rates in all cases setting $\varepsilon = 1$; see Table 8. Note that in this case we have that $\frac{1}{\tau} \in L^q(\Omega)$, $q = \frac{p}{p-1}$; see Assumption 2.6 and Remark 2.7. We solve the problem using $k = 1$ and $k = 2$ polynomial spaces and utilizing both Newton iterative solvers: residual-based line-search (see Algorithm 3.2) and error-oriented (see Algorithm 3.3). In Figure 4, we plot the numerical solution for $\gamma = 0.8$, computed using $p = 1.5$ and $k = 1$.

In Table 9, we present the convergence rates r, as well as the maximum number of nonlinear iterative steps. In particular, the column N_{RB} corresponds to the steps of the residual-based method, and the column N_{EO} corresponds to the error-oriented method. In Table 9, we report only one column with convergence rates r, because both Newton iterative methods yield the same result. We observe that the rates related to $\gamma = 0.72$, $l = 2$, $p = 1.5$ are slightly below the expected value $r = 1$ (see Table 8), but are progressively increased as we use finer meshes. For the $p = 1.8$ test

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TABLE 8
Example 3. *The expected convergence rates r for several values of the parameters γ and p.*

	$u \in W^{l,p}(\Omega)$ with $u = \lvert x\rvert^{\gamma}$, $\varepsilon = 1$		
space $\mathbb{Q}_k$	$\gamma = 0.72, l = 2, p = 1.5$	$\gamma = 0.92, l = 2, p = 1.8$	$\gamma = 1.4, l = 2, p = 3$
	expected rates r		
$k = 1$ and $k = 2$	1	1	1
	$\gamma = 1.72, l = 3, p = 1.5$	$\gamma = 1.92, l = 3, p = 1.8$	$\gamma = 2.4, l = 3, p = 3$
	expected rates r		
$k = 1$	1	1	1
$k = 2$	2	2	2

case, the rates r corresponding to $k = 1$ and $k = 2$ solutions are very close to the expected values. For the last $p = 3$ test case, the rates r corresponding to $k = 1$ are in good agreement with Table 8, but the rates r corresponding to $k = 2$ are a little higher. Here, we would like to note that in several numerical experiments that we have performed using $p > 2$, the numerical rates have been found to be greater than those predicted by the theory, e.g., see Tables 9, 10, and 12 below. Similar rate behavior has been observed in the corresponding tests presented in [17].[3]

Concerning the iterative methods, we observe that both have mesh independent and p independent behavior. However, the performance of the error-oriented method needs fewer steps. However, this does not entail a remarkable reduction of the corresponding CPU times; see the corresponding row in Table 8. The reason is that the error-oriented method is more expensive than the residual-based method, because we solve the linearized system two times per step; see Algorithm 3.3. Continuing with the results for $l = 3$ that are displayed in the last rows in Table 9, we can see that for every p-case the rates r related to $k = 1$ solutions are optimal. For $p = 1.5$, the rates related to $k = 2$ spaces are a little below the expected value. For the test case $p = 1.8$ the rates are close to the expected value and, finally, for the case $p = 3$ the rates become optimal. In conclusion, the results presented in Table 9 show that the approximation error either converges to the predicted or, rate in some cases, related to $\gamma = 0.72$, tends to convergence. Both Newton iterative schemes can be utilized for solving the final nonlinear algebraic system.

4.2. Augmented Lagrangian methods. In this paragraph we present two numerical examples that have been solved using the two ALG1 variants; see Table 1 in section 3.4. Concretely, we have solved the same problems as in Example 2 and in Example 3 in order to compare the residual-based Newton method with the two ALG1 approaches.

Example 4 *(existence of $x_0 \in \Omega : |\nabla u(x_0)| = 0$).* We set $\Omega = (0,1)^2$ and the boundary data u_D and f are then prescribed by the exact solution $u(x,y) = \sin(2\pi(x+y))$, see also Example 2 above. We consider six levels of mesh refinement and four values for $p \in \{1.01, 1.5, 3, 4.3\}$. The numerical tests have been performed using the $k = 1$ polynomial space and $\varepsilon = 1$. We mention that we also performed computations using the remaining values of ε in Table 7. The derived results were similar to those which are described in Table 7 and thus we report only the case $\varepsilon = 1$. In Table 11, we show the numerical convergence rates and the corresponding iterative steps that we have found for the sALG1 method. In Table 10, we show the

[3]For a further investigation, we have performed more computations using different values for γ and p. Again, in some cases, we found rates a little higher than the rates expected by the theory.

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TABLE 9
Example 3: the numerical convergence rates r and the nonlinear iterative steps N for the several values of γ and p.

	$u \in W^{l,p}(\Omega)$ with $u = \lvert x\rvert^{\gamma}$, $\varepsilon = 1$																	
	$\gamma = 0.72, l = 2, p = 1.5$						$\gamma = 0.92, l = 2, p = 1.8$						$\gamma = 1.4, l = 2, p = 3$					
	$k = 1$			$k = 2$			$k = 1$			$k = 2$			$k = 1$			$k = 2$		
$\frac{h}{2^s}$	r	N_{RB}	N_{EO}	r	N_{RB}	N_{EO}	r	N_{RB}	N_{EO}	r	N_{RB}	N_{EO}	r	N_{RB}	N_{EO}	r	N_{RB}	N_{EO}
s =0	-	7	4	-	7	5	-	5	3	-	5	4	-	6	4	-	7	6
s =1	0.69	4	2	0.81	4	2	0.84	3	2	0.99	3	2	0.98	4	2	1.43	3	2
s =2	0.73	4	2	0.82	4	2	0.87	3	2	0.99	3	2	0.99	4	2	1.42	3	1
s =3	0.77	4	2	0.82	4	2	0.89	3	2	0.99	3	2	0.99	3	2	1.42	3	1
s =4	0.78	4	2	0.88	4	2	0.90	3	2	0.99	3	2	0.99	3	2	1.41	3	1
CPU	-	27s	25s	-	130s	120s	-	21s	25s	-	100s	120s	-	22s	19s	-	110s	87s
	$\gamma = 1.72, l = 3, p = 1.5$						$\gamma = 1.92, l = 3, p = 1.8$						$\gamma = 2.4, l = 3, p = 3$					
	$k = 1$			$k = 2$			$k = 1$			$k = 2$			$k = 1$			$k = 2$		
$\frac{h}{2^s}$	r	N_{RB}	N_{EO}	r	N_{RB}	N_{EO}	r	N_{RB}	N_{EO}	r	N_{RB}	N_{EO}	r	N_{RB}	N_{EO}	r	N_{RB}	N_{EO}
s =0	-	5	4	-	5	4	-	4	3	-	5	3	-	5	4	-	7	5
s =1	1	3	2	1.74	3	1	1	3	2	1.85	2	1	1	4	2	2	2	1
s =2	1	3	2	1.76	2	1	1	3	2	1.87	2	1	1	3	2	2	2	1
s =3	1	3	2	1.77	2	1	1	3	1	1.90	2	1	1	3	2	2	2	1
s =4	1	3	1	1.78	2	1	1	2	1	1.91	2	1	1	3	2	2	2	1
CPU	-	17s	19s	-	76s	46s	-	17s	20s	-	51s	45s	-	23s	21s	-	49s	54s

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TABLE 10
Example 4. *mALG*1 *method. The convergence rates and the iterative steps for several values of p.*

-	sALG1, $\varepsilon = 1$ and $u_h^0 = P(u_{2h}^n)$ and $k = 1$							
grid size	$p = 1.01$		$p = 1.5$		$p = 3$		$p = 4.3$	
$\frac{h}{2^s}$	r	N	r	N	r	N	r	N
s=0	-	4	-	4	-	4	-	3
s=1	1.1	3	1.1	4	1.2	4	1.4	3
s=2	1.0	3	1.3	3	1.1	3	1.4	2
s=3	1.3	2	1.2	3	1.1	3	1.3	2
s=4	1.2	2	1.1	2	1.0	3	1.2	1

TABLE 11
Example 4. *sALG*1 *method. The convergence rates and the iterative steps for several values of p.*

-	mALG1, $\varepsilon = 1$ and $u_h^0 = P(u_{2h}^n)$ and $k = 1$							
grid size	$p = 1.01$		$p = 1.5$		$p = 3$		$p = 4.3$	
$\frac{h}{2^s}$	r	N	r	N	r	N	r	N
s=0	-	4	-	4	-	7	-	12
s=1	0.86	3	0.98	2	1.1	3	1.2	3
s=2	1.0	3	1.0	2	1.0	2	1.2	2
s=3	1.0	2	1.0	2	1.0	2	1.0	2
s=4	1.0	2	1.0	2	1.0	2	1.0	1

TABLE 12
Example 5. *sALG*1 *for solving problems with low regularity solution. The numerical convergence rates and the iterative steps for several values of* γ *and p.*

	sALG1, $u \in W^{l,p}(\Omega)$ with $u = \lvert x\rvert^\gamma$, $\varepsilon = 1$											
	$\gamma = 0.72, l = 2, p = 1.5$				$\gamma = 0.92, l = 2, p = 1.8$				$\gamma = 1.4, l = 2, p = 3$			
$\frac{h}{2^s}$	$k = 1$	N	$k = 2$	N	$k = 1$	N	$k = 2$	N	$k = 1$	N	$k = 2$	N
-	Rates r and steps N for sALG1											
$s = 0$	-	22	-	14	-	17	-	11	-	24	-	16
$s = 1$	0.85	18	0.85	10	1.02	16	0.95	11	1.50	16	1.45	11
$s = 2$	0.79	17	0.86	9	0.94	15	0.95	8	1.30	12	1.43	10
$s = 3$	0.79	16	0.86	8	0.93	13	0.96	7	1.22	12	1.42	10
$s = 4$	0.80	14	0.87	7	0.94	11	0.95	6	1.17	9	1.41	7
CPU	-	94s	-	394s	-	120s	-	306s	-	101s	-	440s
	$\gamma = 1.72, l = 3, p = 1.5$				$\gamma = 1.92, l = 3, p = 1.8$				$\gamma = 2.4, l = 3, p = 3$			
$\frac{h}{2^s}$	$k = 1$	N	$k = 2$	N	$k = 1$	N	$k = 2$	N	$k = 1$	N	$k = 2$	N
-	Rates r and steps N for sALG1											
$s = 0$	-	20	-	10	-	18	-	8	-	24	-	14
$s = 1$	1.35	16	1.76	6	1.35	14	1.90	6	1.17	18	2.1	12
$s = 2$	1.46	11	1.77	6	1.39	12	1.91	6	1.28	12	2.07	10
$s = 3$	1.39	8	1.78	6	1.35	8	1.92	6	1.30	11	2.04	10
$s = 4$	1.30	7	1.79	6	1.29	7	1.90	6	1.23	10	2.02	9
CPU	-	64s	-	344s	-	61s	-	389s	-	117s	-	471s

corresponding data of the mALG1 method. In both tables the column associated with N indicates the number of nonlinear (Newtons) iterations necessary to achieve the convergence criteria. In general, we observe that both methods sALG1 and mALG1 converge quite fast, as few as 2 iterations on fine meshes. In some cases, the proposed

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TABLE 13

Example 5. *mALG*1 *for solving problems with low regularity solution. The numerical convergence rates and the iterative steps for several values of* γ *and* p.

	mALG1, $u \in W^{l,p}(\Omega)$ with $u = \lvert x\rvert^{\gamma}$, $\varepsilon = 1$											
	$\gamma = 0.72, l = 2, p = 1.5$				$\gamma = 0.92, l = 2, p = 1.8$				$\gamma = 1.4, l = 2, p = 3$			
$\frac{h}{2^s}$	$k=1$	N	$k=2$	N	$k=1$	N	$k=2$	N	$k=1$	N	$k=2$	N
-	Rates r and steps N for mALG1											
$s=0$	-	9	-	9	-	6		6	-	9	-	10
$s=1$	0.73	6	0.85	9	0.77	8	0.91	5	0.97	5	1.32	10
$s=2$	0.85	6	0.90	5	0.90	8	0.95	7	1.02	7	1.40	6
$s=3$	0.86	5	0.90	5	0.91	7	0.95	6	1.02	5	1.41	4
$s=4$	0.87	5	0.90	4	0.92	7	0.95	6	1.01	4	1.41	3
CPU	-	133s	-	1174s	-	188s	-	1597s	-	98s	-	883s
	$\gamma = 1.72, l = 3, p = 1.5$				$\gamma = 1.92, l = 3, p = 1.8$				$\gamma = 2.4, l = 3, p = 3$			
$\frac{h}{2^s}$	$k=1$	N	$k=2$	N	$k=1$	N	$k=2$	N	$k=1$	N	$k=2$	N
-	Rates r and steps N for mALG1											
$s=0$	-	8	-	8	-	5	-	6	-	9	-	8
$s=1$	1.00	4	1.74	3	1.02	4	1.81	5	0.97	7	2.02	8
$s=2$	1.00	3	1.76	2	1.01	3	1.87	2	1.02	5	2.07	3
$s=3$	1.00	3	1.78	2	1.01	2	1.90	2	1.01	4	2.00	2
$s=4$	1.00	2	1.79	1	1.00	2	1.91	1	1.01	4	2.00	1
CPU	-	76s	-	356s	-	61s	-	335s	-	76s	-	338s

mALG1 method requires fewer steps to reach the stopping criterion; see, for example, the $p = 1.5$ case. The number N of iterations remains stable and very low for the last mesh levels, thus we can say that both ALG1 methods exhibit a mesh independent behavior. Also we observe that the convergence rates of sALG1 appear to be a little higher than the theory predicts. On the other hand the convergence rates r corresponding to the mALG1 method are very close to the rates predicted by the theory.

Example 5 *(low regularity solution).* We consider $\Omega = (-0.5, 0.5)^2$ and the exact solution $u(x, y) = (x^2 + y^2)^{\frac{\gamma}{2}}$; see also Example 3 and Tables 8 and 9. We have solved the problem using linear elements $k = 1$ giving to γ and p the values of the second row in Table 8. We compare the two different ALG1 methods by choosing three different test cases associated with three different values for p, i.e., $p = \{1.5, 1.8, 3\}$. Table 12 corresponds to the sALG1 method and it displays the convergence rates of the error and the iterative steps N for every mesh level. The results related to the mALG1 method are displayed in Table 13. The initial guess is defined to be the projection of the previous coarse mesh solution. We begin with the results of sALG1 displayed in Table 12 which are associated with $l = 2$ regularity. We observe that for $p = 1.5$ the rates, for both $k = 1$ and $k = 2$ spaces, are slightly less than the expected optimal $r = 1$ rate, but they progressively increase towards the optimal value. For the case of $p = 1.8$ the rates r of both numerical solutions $k = 1$ and $k = 2$ are very close to the optimal value $r = 1$ which is determined by the regularity of u. On the other hand the rates for the $p = 3$ case are a little higher on the first meshes but are progressively reduced. In general, the convergence rates in Table 12, which are related to $l = 2$ regularity, are closer to the rates presented in Table 8, in comparison to the rates that we show in Table 9. Looking at Table 12, the column with the number of nonlinear iterative steps, we observe that in general the sALG1 method has a mesh independent behavior. The iterative steps corresponding to $k = 1$ solutions are considerably more than the steps corresponding to $k = 2$ solutions. This does not have an impact on the total CPU

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time, since we observe that the CPU time of $k = 2$ computations is much higher that the CPU time of $k = 1$ computations; see the corresponding row in Table 12.

Also, Table 12 shows the results for the $l = 3$ tests (see last rows). We observe that the rates related to $k = 1$ for all p-cases are a little higher than the $r = 1$ (expected value). On the other hand the rates r related to $p = 1.5$, $k = 2$ solutions are a little lower than expected. For the two other p-cases, the convergence rates of $k = 2$ numerical solutions are close to the expected $r = 2$ rate. Concerning the variations of the numbers N of the iterative steps, we observe similar behavior to the previous $l = 2$ test case. More precisely, although the steps of the $k = 1$ numerical solutions are again much higher than the steps of the $k = 2$ solutions, the corresponding CPU time of $k = 1$ solutions is smaller. This occurs due to the fact that ALG1 methods introduce two new variables; see Table 1. Thus, when we use $k = 2$ spaces, the resulting algebraic linear system at each Newton step has a high number of unknowns, and therefore its solution requires much more computing time. Here, advanced techniques, e.g., efficient preconditioners such as, for instance, multigrid techniques must be applied. This is a topic that we will discuss in a forthcoming paper; see also Remark 3.8.

We continue with the results of mALG1 method for the $l = 2$ regularity case; see first rows in Table 13. For the $p = 1.5$ test case, we observe that the rates r corresponding to $k = 1$ solutions are a little lower than the optimal value but, moving to finer meshes, the rates are approaching the optimal value. For the other two p-cases the rates of $k = 1$ solutions are much closer to the expected $r = 1$ value. Concerning the rates of $k = 2$ solutions, we can see that they are close to $r = 1$ which is not surprising since $u \in W^{l=2,p}$. We have to remark that the rates obtained for the $p = 3$ test case are a little bit higher than expected. Finally, we mention that, for all p-cases, the required number of iterative steps is not increased as we use finer grids, even for $k = 2$ spaces. This indicates that the proposed mALG1 method follows a mesh independent behavior. Here, we also observe that the number of iterative steps for both cases $k = 1$ and $k = 2$ are very close, and such a behavior did not appear in the sALG1 method. Concerning the CPU times displayed in the last row, similar comments to previous tests of the sALG1 method can be expressed.

The last rows in Table 13 show the results related to the $l = 3$ regularity test case. We observe that for each of the three p-tests the error related to $k = 1$ spaces converges at the optimal rate $r = 1$, as predicted by Theorem 2.12; see also Table 8. As in the previous examples, the error convergence rates related to $k = 2$ numerical solutions are a little below the expected value for the case of $p = 1.5$, but the rates are in agreement with the predicted rates for the rest of the p-cases. In general, we can say that the rates behave in a manner similar to Example 3, where the residual-based method applied; see the last lines in Table 8. Also, Table 13 presents the actual number of nonlinear iterative steps at every mesh. We see that the iterative steps of both $k = 1$ and $k = 2$ solutions are very close. For each choice of p, the solution is obtained in very few iterative steps even on the last mesh levels. Also we note that the number of iterative steps is not increased as we increase the value of p. We did not observe the same in Table 12 which shows the results for the sALG1 method. This fact indicates the robustness of the new proposed mALG1 method with respect to the choice of p. Finally, in the last row in Table 13, we can see the corresponding CPU times. As in the $l = 2$ case and as in the sALG1 method, the CPU times of $k = 2$ solutions are much larger than the CPU times of $k = 1$ solutions. This is not the case for the two Newton methods; see the last row in Table 9.

As a final comment we can say that the two ALG1 methods appear to be quite appropriate for solving the nonlinear system in (3.4). This seems to be a result of the

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particular decomposition of the problem and the treatment of the nonlinear terms. On the other hand, this ALG1 decomposition introduces additional unknowns, which results in the creation of a large system.

5. Conclusions. In this article, we applied finite element methods for the numerical solution of p-power-law diffusion problems. Under some regularity assumptions for the exact solution we derived a priori error estimates. The theoretical error estimates were demonstrated by numerical examples using several values for the parameters of the problem. Next, we presented nonlinear iterative solvers. Initially, we analyzed two Newton-like iterative methods, a residual-based and an error-oriented techniques. We discussed in detail the condition properties of the Newton iterative matrix. Even though the error-oriented procedure finds the solution in fewer iterations than the residual-based technique, the latter is easier to actualize. However, in the error-oriented method the linear system is solved twice, which finally makes the two methods appear to be equivalent in terms of CPU time. In addition, we applied two ALG methodologies for solving the original p-type elliptic problem. The resulting iterative methods can be characterized by the simplicity of the algorithm and are able to provide a nice handling of the nonlinear terms via the decomposition of the original problem. Nevertheless, the introduction of the additional unknowns that accompanies the ALG1 framework, yields a large system at each Newton step. For an efficient solution of this system, it seems that we need to apply more advanced techniques, which we envisage to develop in future work.

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